Database Management Systems (DBMS)
GTU # 3130703

Unit-2

Data Models

Dr.Ramesh
CE/IT Department

Shree Swaminarayan Institute of Technology

Outline

- Basic concept of E-R diagram
- Types of Attributes
- Mapping Cardinality
- Weak Entity Sets
- Extended E-R features
- Generalization and Specialization
- Constraints on Specialization and Generalization
- Aggregation
- E-R diagram of Hospital Management System
- Reduction to E-R Database Schema
- Database Models
- Integrity Constraints

Basic concept of E-R diagram

Section - 1

Basic concepts

- What is Database Design?
  - Database Design is a collection of processes that facilitate the **designing**, **development**, **implementation** and **maintenance** of enterprise database management systems.
- What is E-R diagram?
  - E-R diagram: (Entity-Relationship diagram)
  - It is **graphical (pictorial) representation** of database.
  - It uses different types of symbols to represent different objects of database.

Entity

- An entity is a **person**, a **place** or an **object**.
- An entity is represented by a **rectangle** which contains the name of an entity.
- Entities of a college database are:
  - Student
  - Professor/Faculty
  - Course
  - Department
  - Result
  - Class
  - Subject

Entity Name

Symbol

Student

Faculty

Course

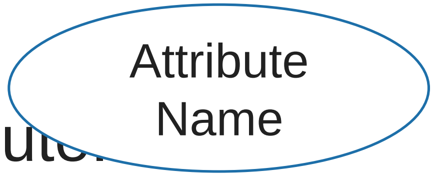
Exercise	Write down the different entities of bank database.
Exercise	Write down the different entities of hospital database.

Entity Set

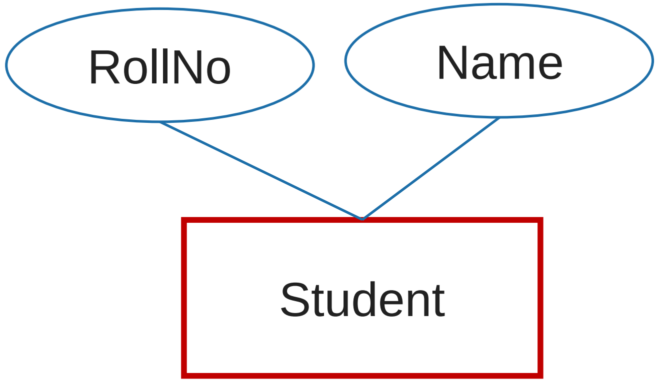
- It is a **set (group) of entities** of **same type**.
- Examples:
  - All persons having an account in a bank
  - All the students studying in a college
  - All the professors working in a college
  - Set of all accounts in a bank

Attributes

- Attribute is **properties** or details about an entity.
- An attribute is represented by an **oval** containing name of an attribute.
- Attributes of Student are:
  - Roll No
  - Student Name
  - Branch
  - Semester
  - Address
  - Mobile No
  - Age
  - SPI
  - Backlogs



Symbol

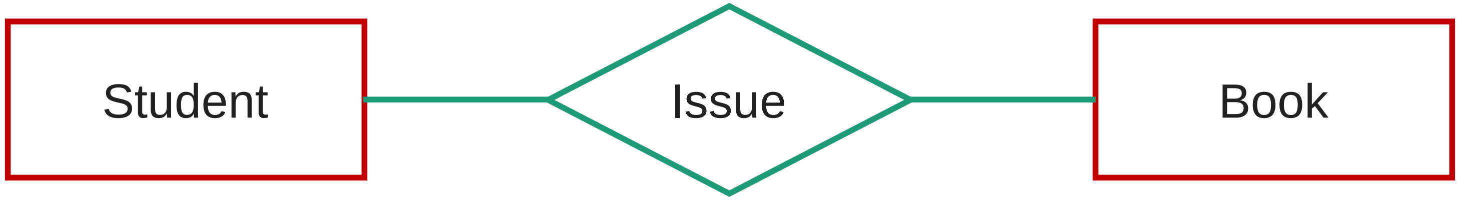


Exercise	Write down the different attributes of Faculty entity.
Exercise	Write down the different attributes of Account entity.

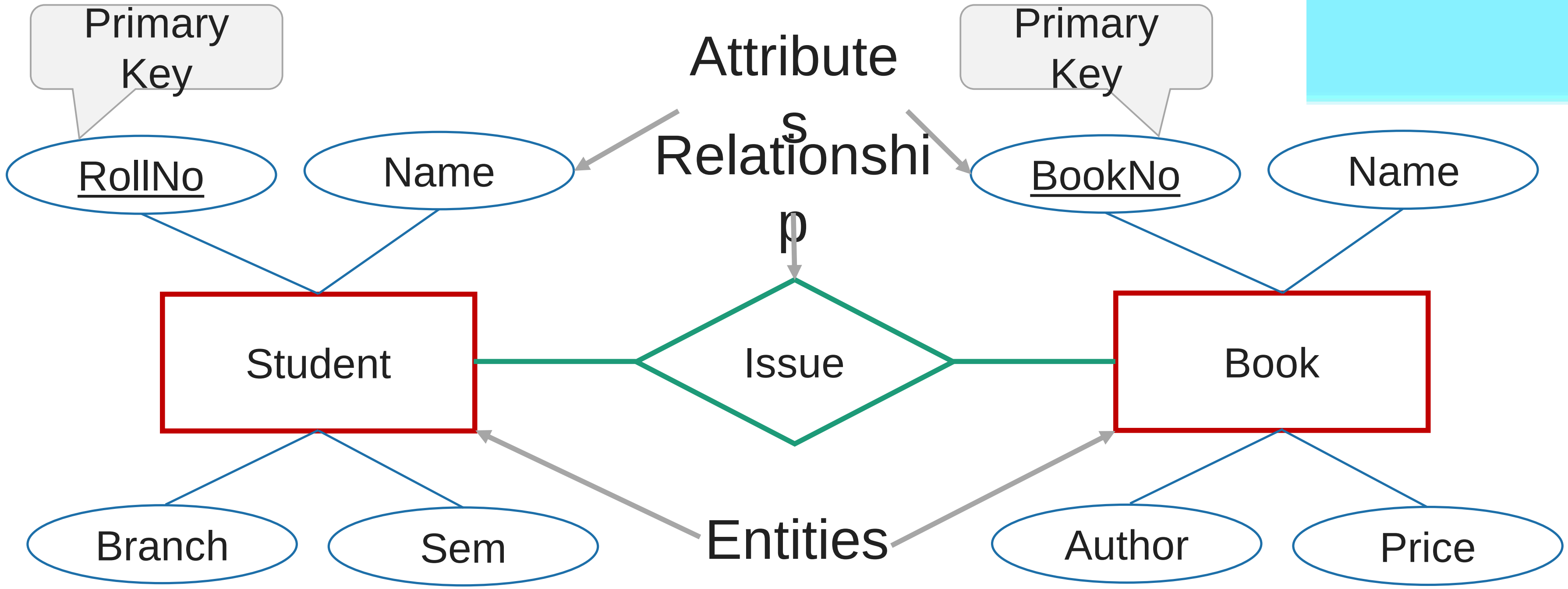
Relationship

- Relationship is an **association** (connection) between several entities.
- It should be placed between two entities and a line connecting it to an entity.
- A relationship is represented by a **diamond** containing relationship's name.

Symbol

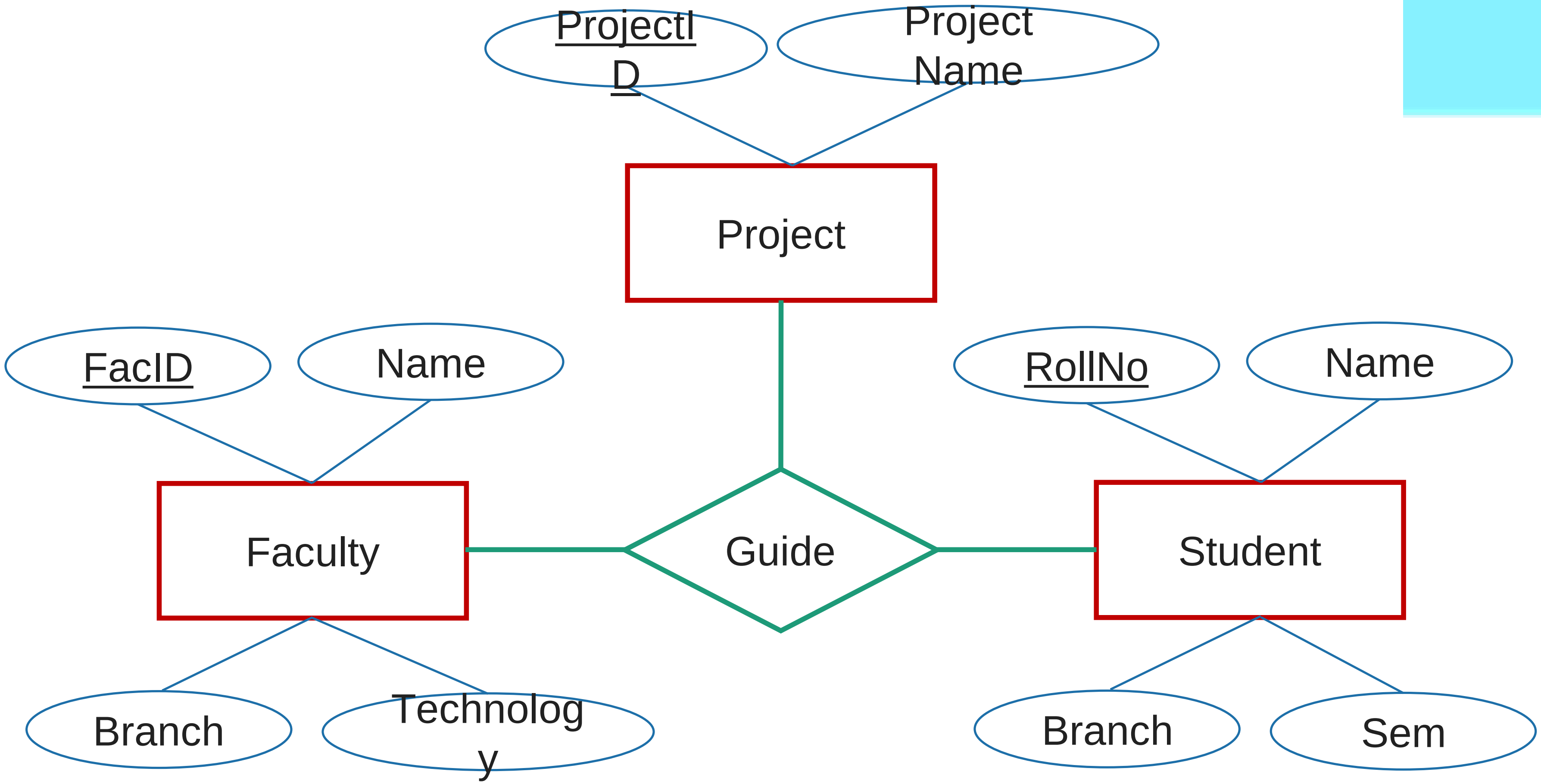


E-R Diagram of a Library System



Each and every entity must have one primary key attribute.
Relationship between 2 entities is called binary relationship

Ternary Relationship



Relationship between 3 entities is called ternary relationship.

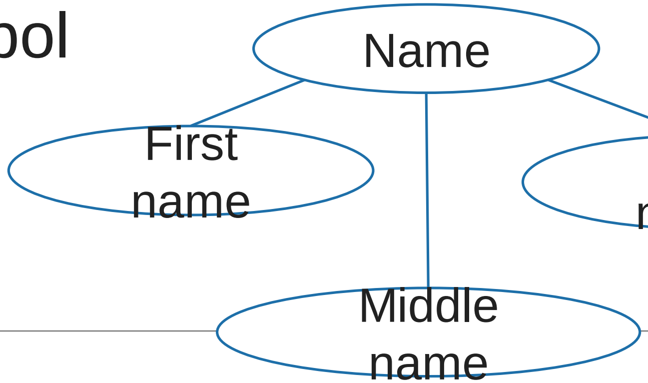
Exercise

- Draw an E-R diagram of following pair of entities
  - Customer & Account
  - Customer & Loan
  - Doctor & Patient
  - Student & Project
  - Student & Teacher
    - Note: Take four attributes per entity with one primary key attribute. Keep proper relationship between two entities.

Types of Attributes

Section - 2

Types of Attributes

Simple Attribute	Composite Attribute
Cannot be divided into subparts	Can be divided into subparts
E.g. RollNo, CPI	E.g. Name (first name, middle name, last name) Address
Symbol 	Symbol 

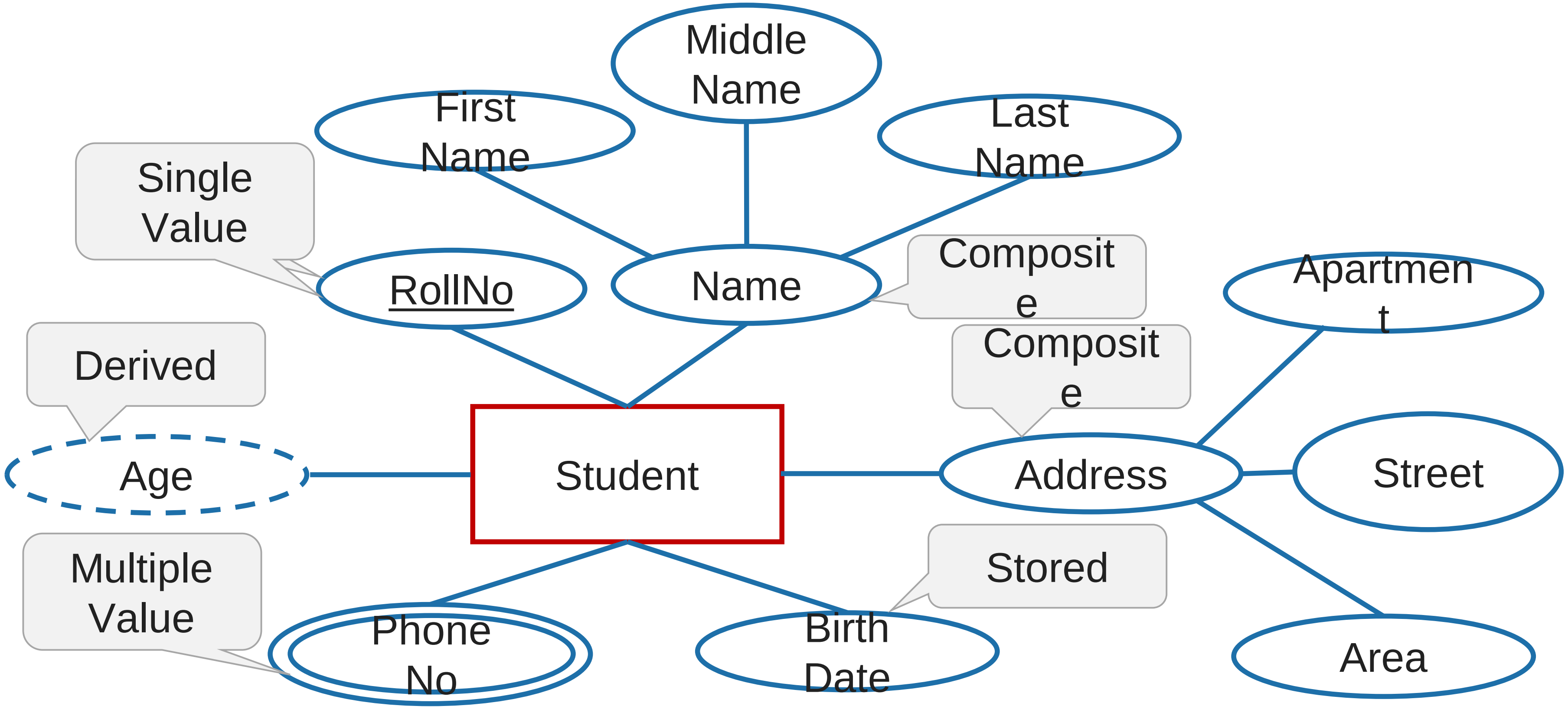
Types of Attributes

Single-valued Attribute	Multi-valued Attribute
Has single value	Has multiple (more than one) value
E.g. RollNo, CPI	E.g. PhoneNo (person may have multiple phone nos) EmailID
Symbol 	Symbol 

Types of Attributes

Stored Attribute	Derived Attribute
It's value is stored manually in database	It's value is derived or calculated from other attributes
E.g. Birthdate	E.g. Age (can be calculated using current date and birthdate)
Symbol 	Symbol 

Entity with all types of Attributes



Exercise

- Draw an E-R diagram of Banking Management System.
- Draw an E-R diagram of Hospital Management System.
- Draw an E-R diagram of College Management System.
  - Take only 2 entities
  - Keep proper relationship between two entities
  - Use all types of attributes

❖ Relationships

- Relationship → association among 2 or more entities.
- Ex: teacher teaches student

  ↓ teaches: relationship

★ Degree of Relationship: Denotes the number of entity types tha
participate in a relationship.

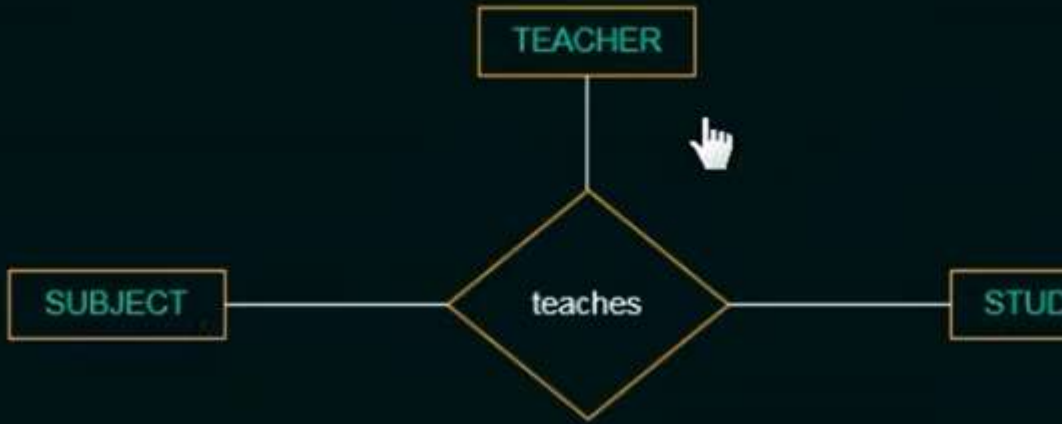
1. Unary relationship:
   ➢ Exists when there is an association with only one entity.

❖ Relationships

★ Degree of Relationship:

2. Binary relationship:
   - ➢ Exists when there is an association among two entities.

3. Ternary relationship:
   - ➢ Exists when there is an association among three entities.



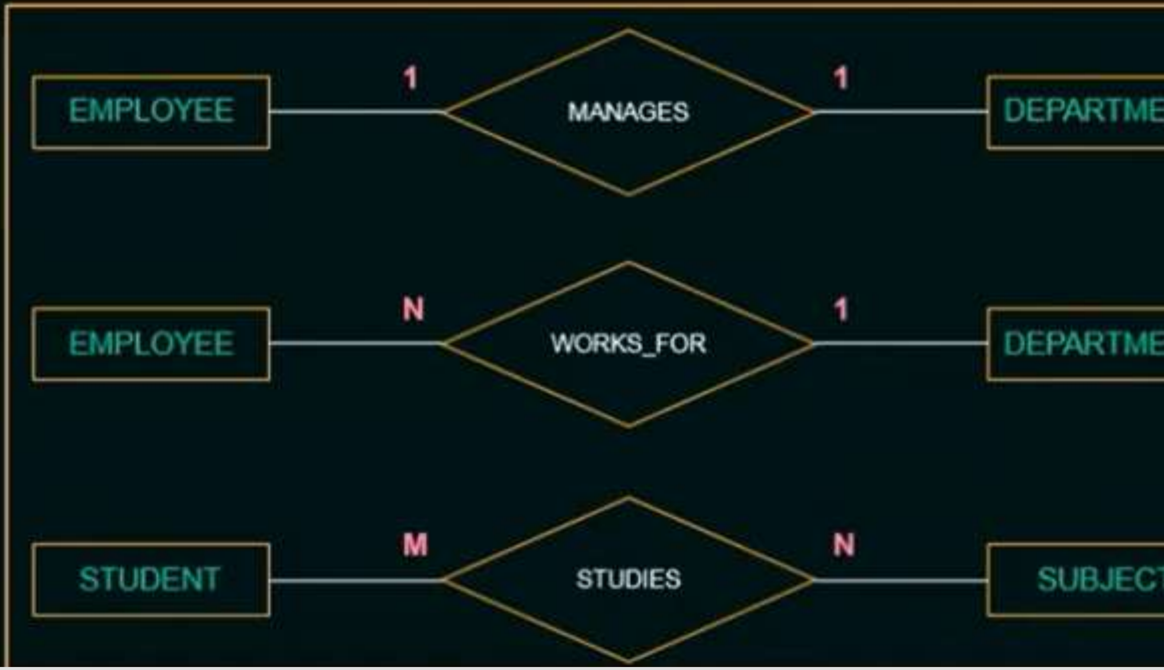
❖ Relationships

★ Relationship Constraints:

1. Cardinality Ratio

➢ Maximum number of relationship instances that an entity can participate in.

➢ Possible cardinality ratios for binary relationship → 1:1, 1:N, N:1, M:N.



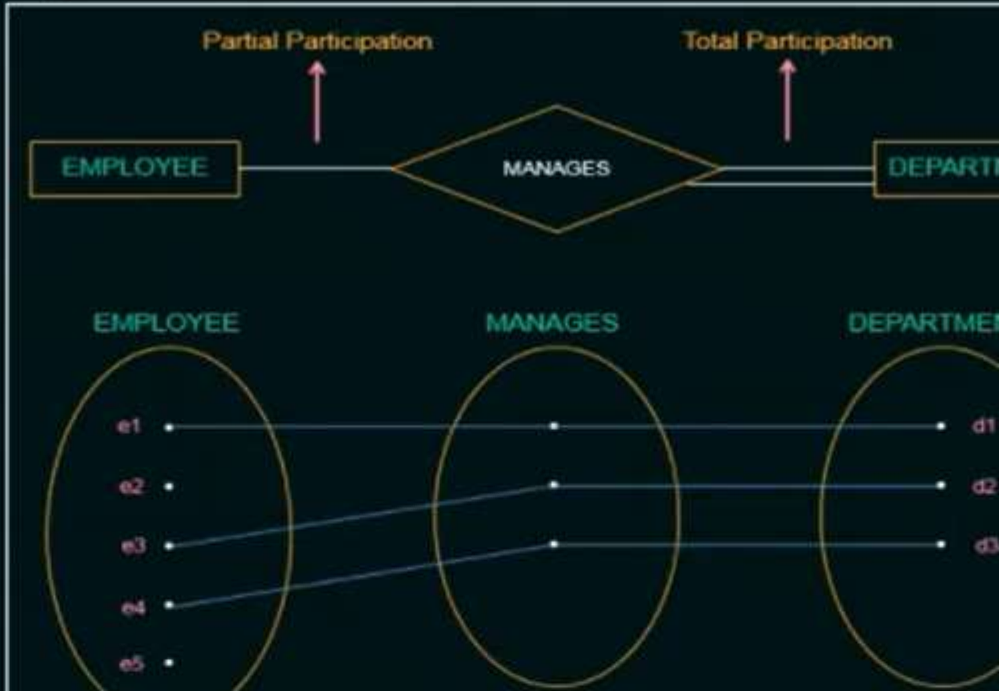
❖ Relationships

★ Relationship Constraints:

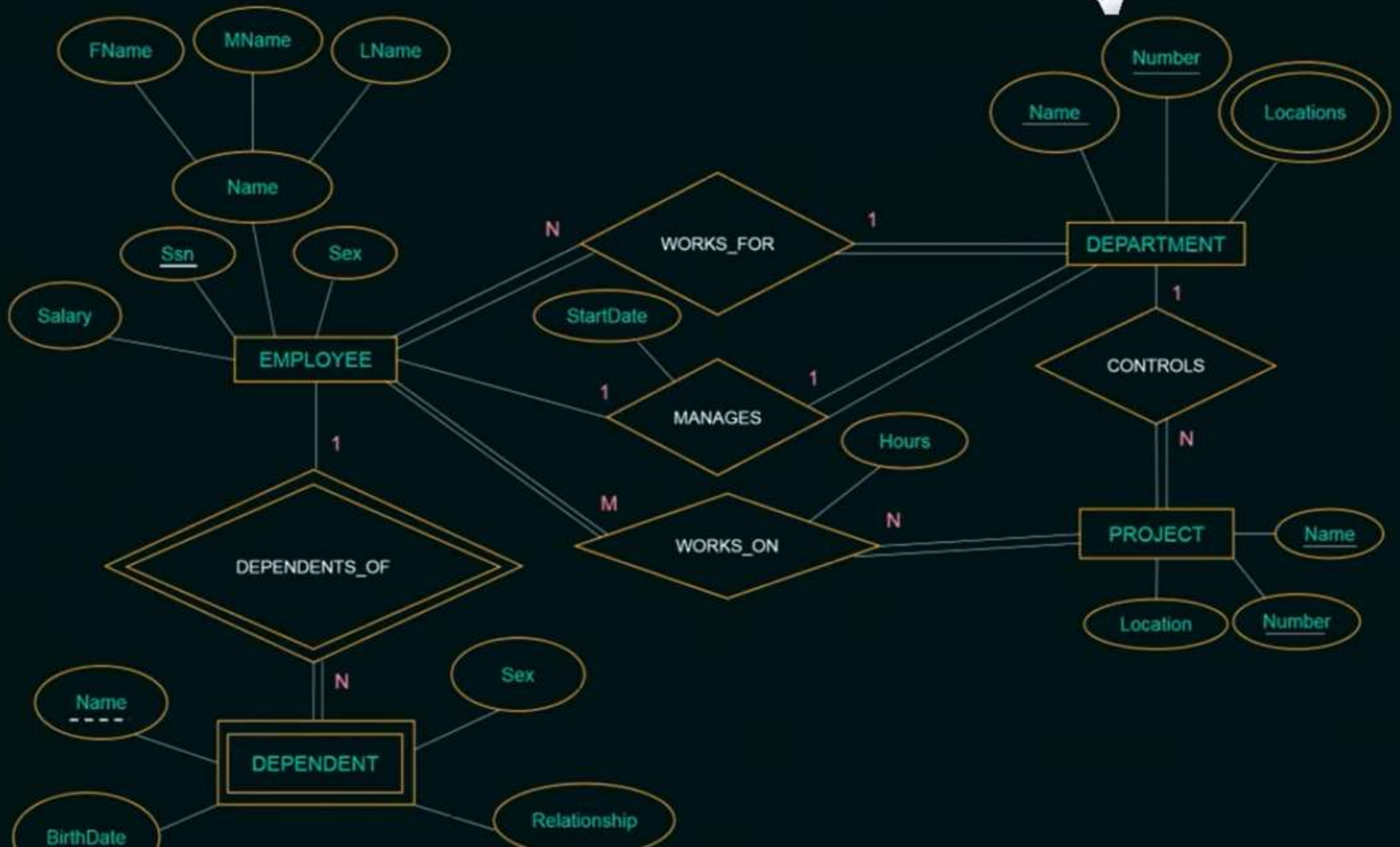
2. Participation Constraints

➢ Specifies whether existence of an entity depends on its being related to another entity.

➢ 2 types: Total participation & Partial participation.



❖ ER Diagram for COMPANY Database



Relationships

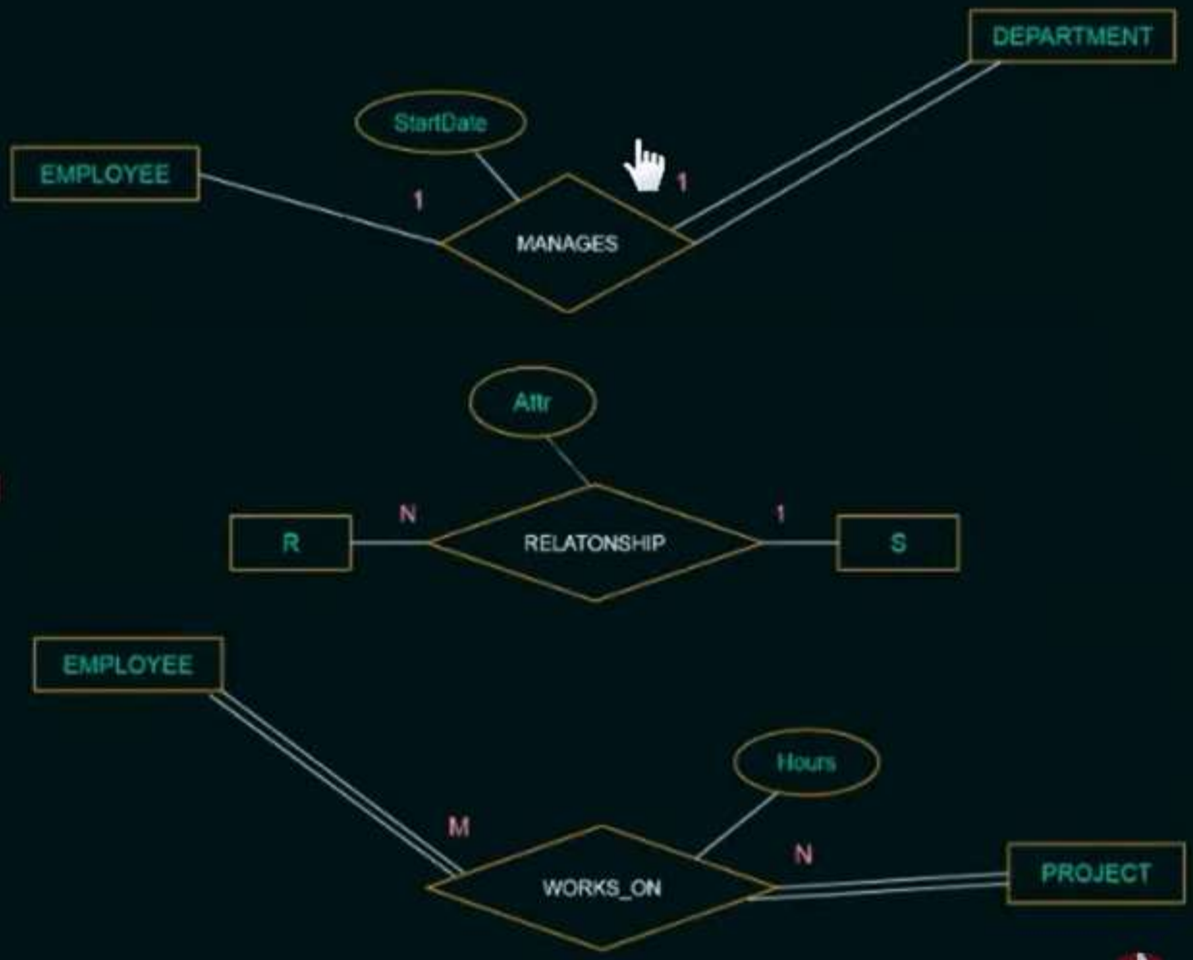
Attributes of Relationship Types

- Attributes of 1:1 or 1:N relationship types can be migrated to one of the participating entity types.
  - In 1:1 relationship type, attributes can be migrated to either of the entity types.
  - In 1:N or N:1 relationship type, attributes are migrated only to the entity type on the N-side of the relationship.

❖ Relationships

★ Attributes of Relationship Types

- In M:N relationship type, some attributes can be determined by a combination of participating entities.



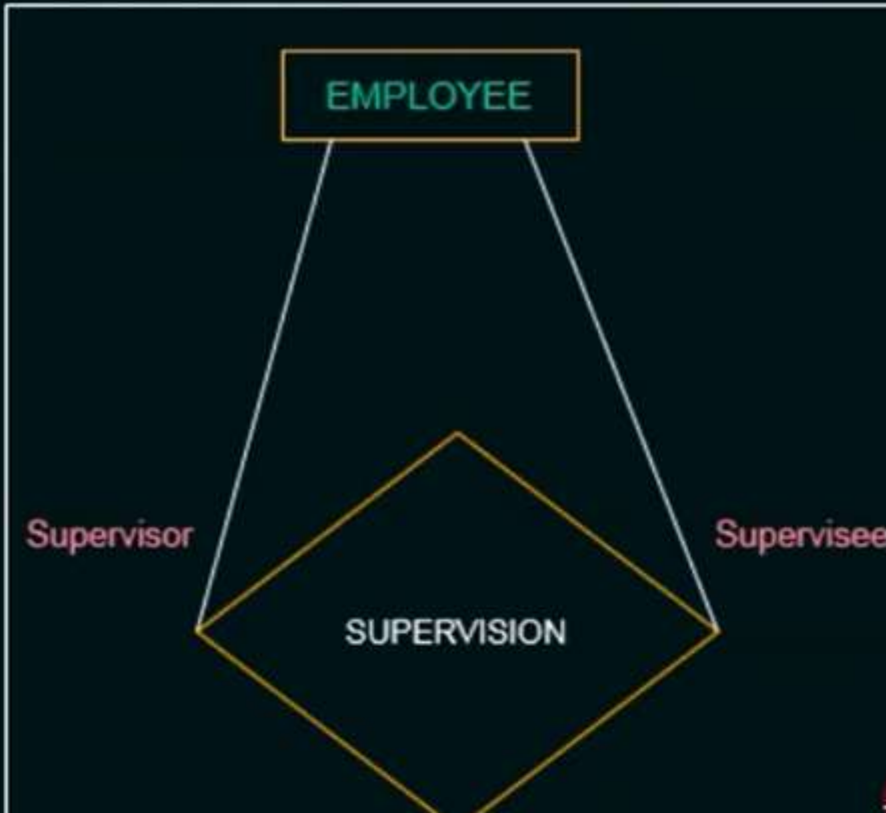
❖ Relationships

★ Role Names

- Signifies the role that a participating entity plays in each relationship instance.

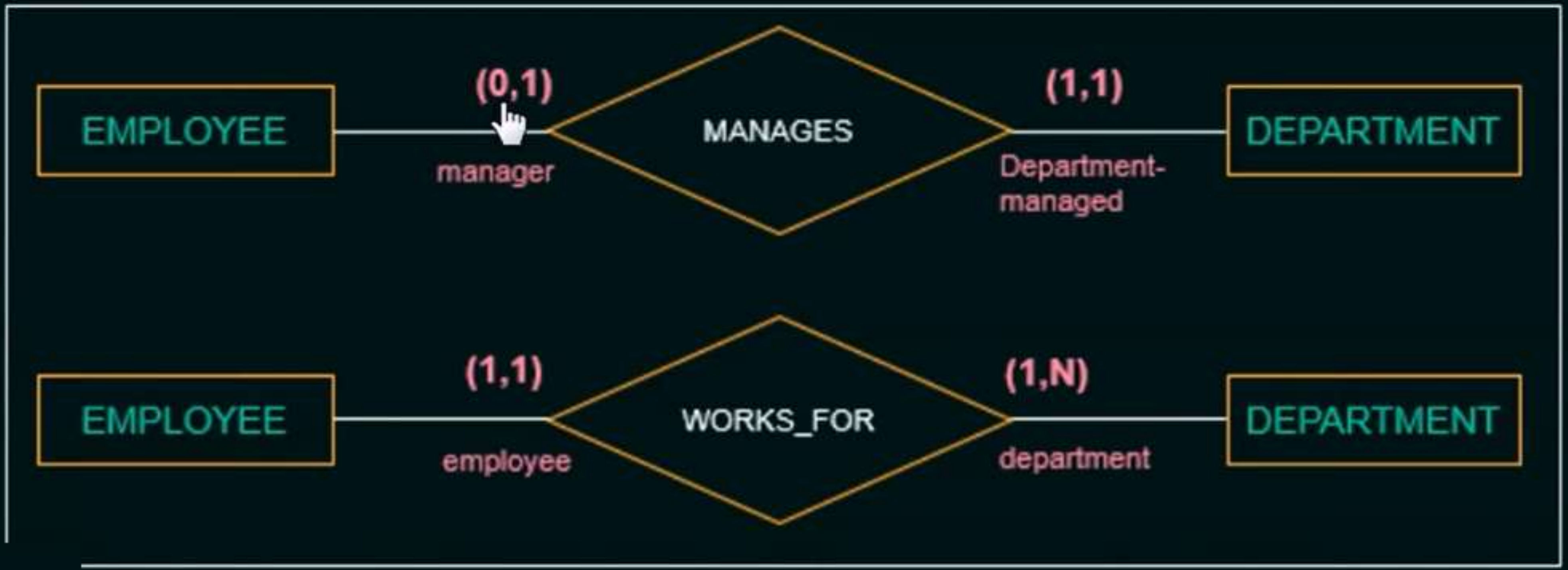
★ Recursive Relationships

- Same entity type participates more than once in a relationship type in different roles.



❖ Alternative Notations for ER Diagrams

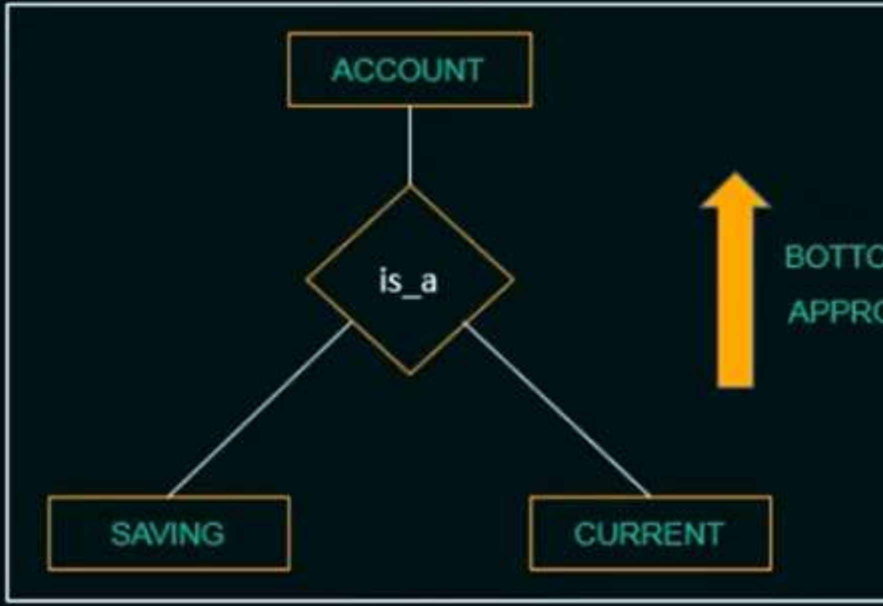
- Associates a pair of integer numbers (min, max) with each participation of an entity type in a relationship type, where $0 \le min \le max$ and $max \ge 1$.



❖ Enhanced ER Model

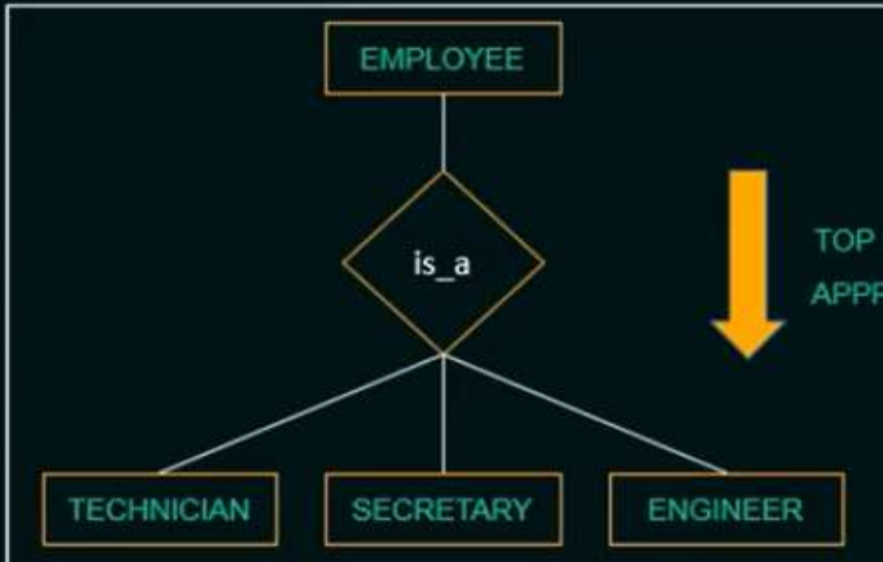
★ **Generalization:**

- **Bottom-up approach** where two lower level entities combine to form a higher level entity.



★ **Specialization:**

- **Top-down approach** where it defines set of subclasses of an entity type.



Exercise

- Draw an E-R diagram of Banking Management System.
- Draw an E-R diagram of Hospital Management System.
- Draw an E-R diagram of College Management System.
  - Take only 2 entities
  - Keep proper relationship between two entities
  - Use all types of attributes

Exercise

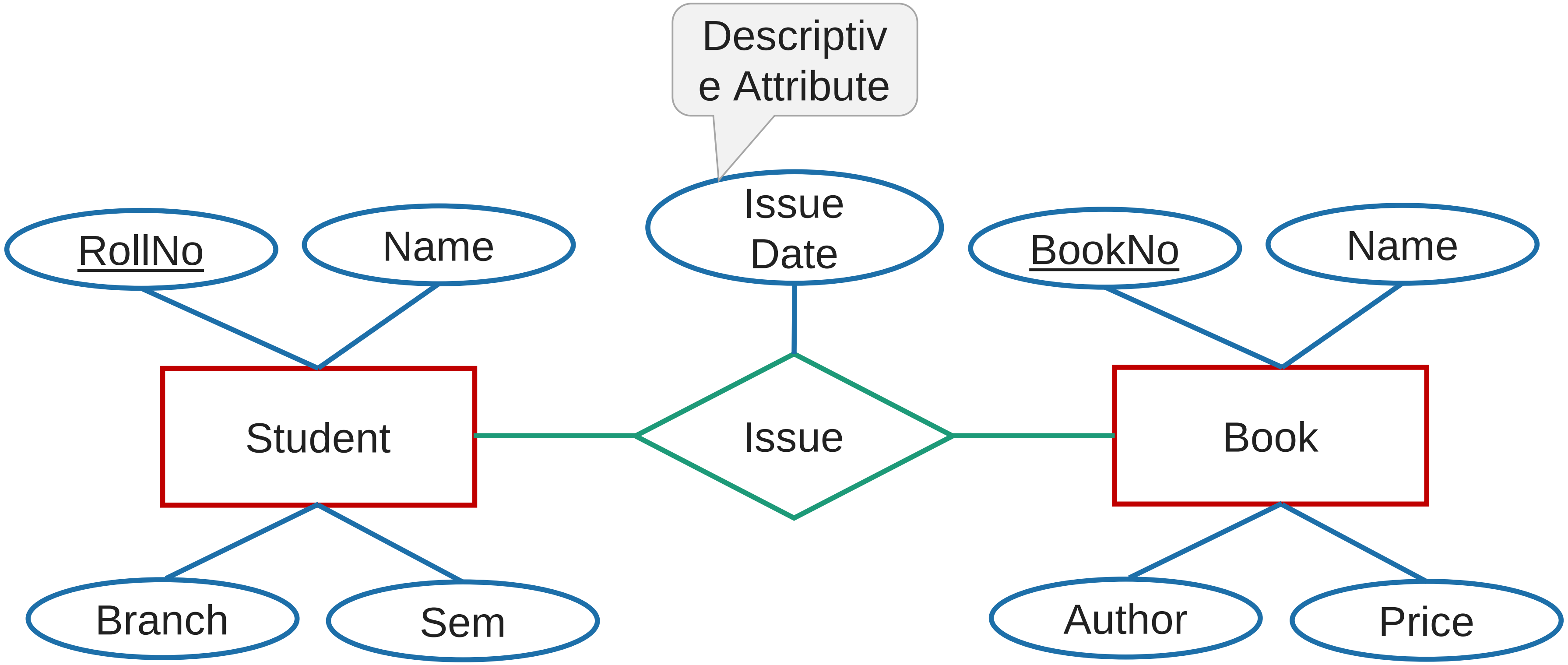
- Draw an E-R diagram of Banking Management System.
- Draw an E-R diagram of Hospital Management System.
- Draw an E-R diagram of College Management System.
    - Take only 2 entities
    - Keep proper relationship between two entities
    - Use all types of attributes

Exercise

- Draw an E-R diagram of Banking Management System.
- Draw an E-R diagram of Hospital Management System.
- Draw an E-R diagram of College Management System.
  - Take only 2 entities
  - Keep proper relationship between two entities
  - Use all types of attributes

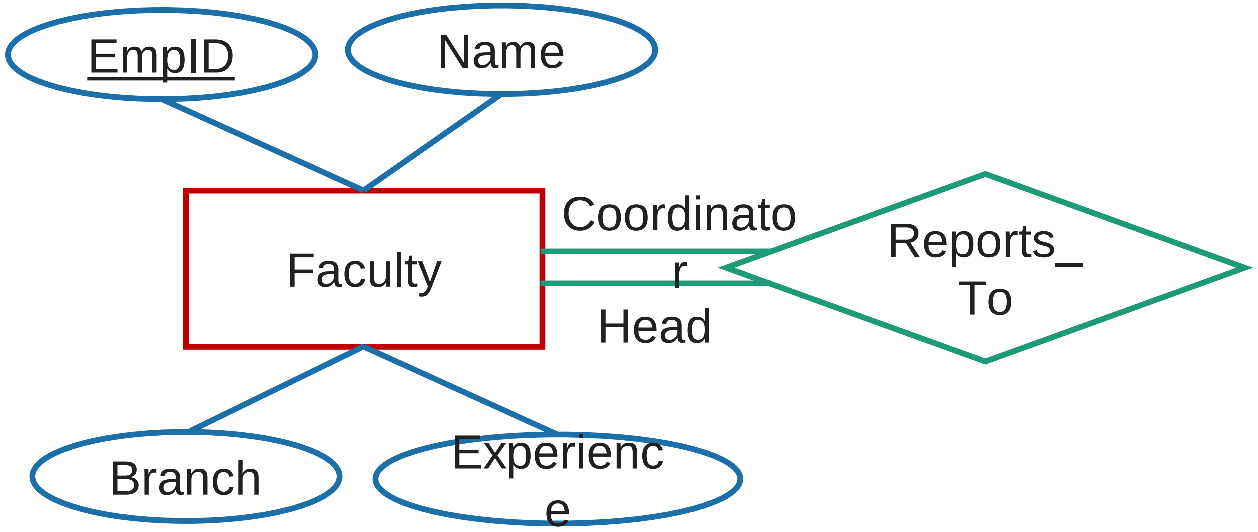
Descriptive Attribute

- **Attributes of the relationship** is called descriptive attribute.



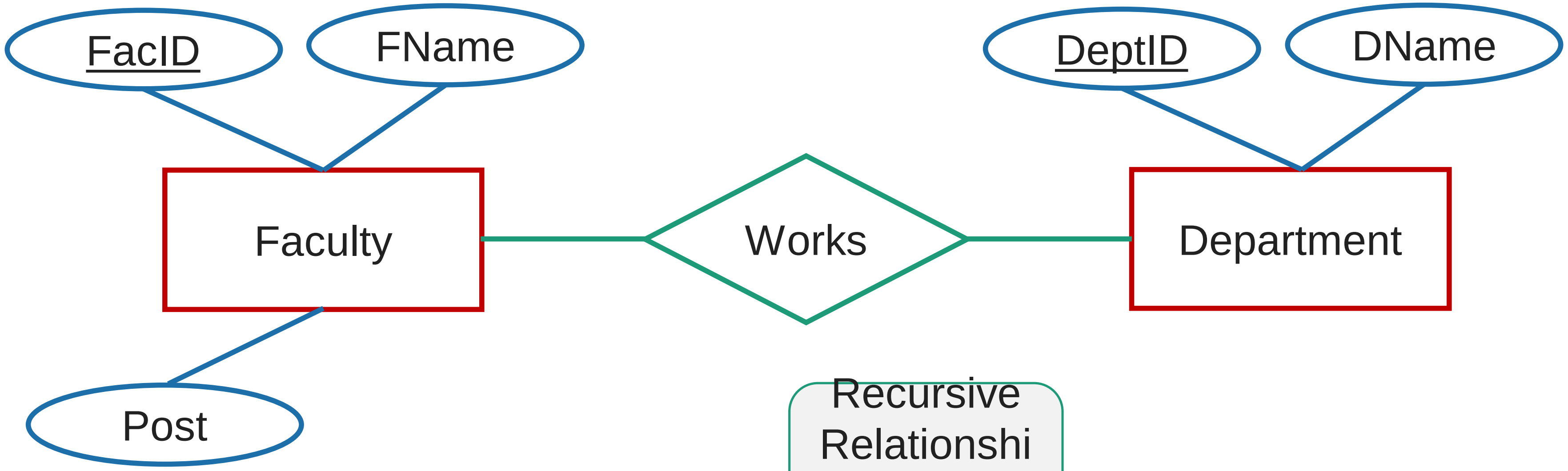
Role

- Roles are indicated by labeling the lines that connect diamonds (relationship) to rectangles (entity).
- The labels “Coordinator” and “Head” are called roles; they specify Faculty entities interact with whom via Reports_To relationship set.
- Role labels are optional, and are used to clarify semantics (meaning) of the relationship.

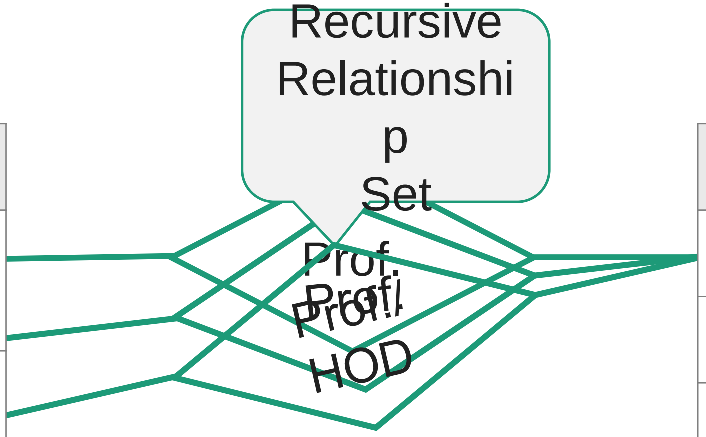


Recursive Relationship Set

- The same **entity participates in a relationship set more than once** then it is called recursive relationship set.



FName	Post
Ajay	Professor
Haresh	Professor
Ramesh	HOD



DName
Computer
Civil
Mechanical

Mapping Cardinality

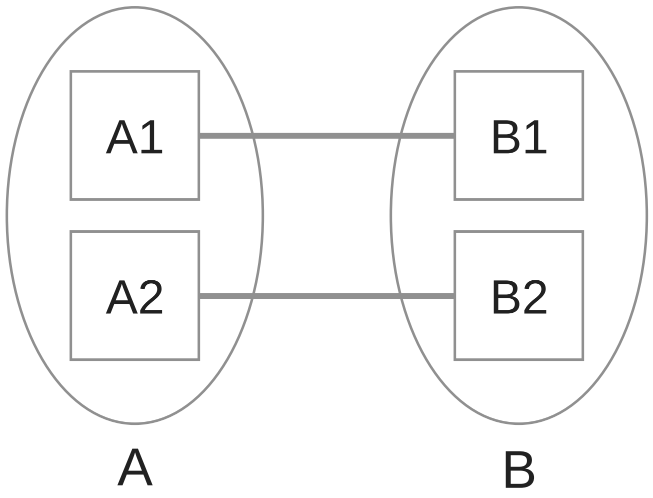
Section - 3

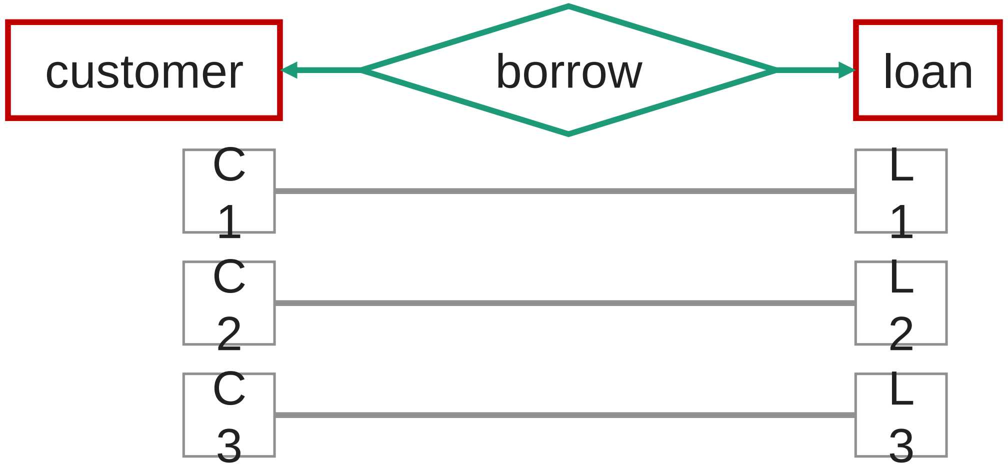
Mapping Cardinality (Cardinality Constraints)

- It represents the **number of entities of another entity set** which are **connected to an entity** using a relationship set.
- It is most **useful in describing binary relationship sets**.
- For a binary relationship set the mapping cardinality must be one of the following types:
  - One to One
  - One to Many
  - Many to One
  - Many to Many

One-to-One relationship (1 – 1)

- An entity in **A is associated with only one entity in B** and an entity in **B is associated with only one entity in A**.

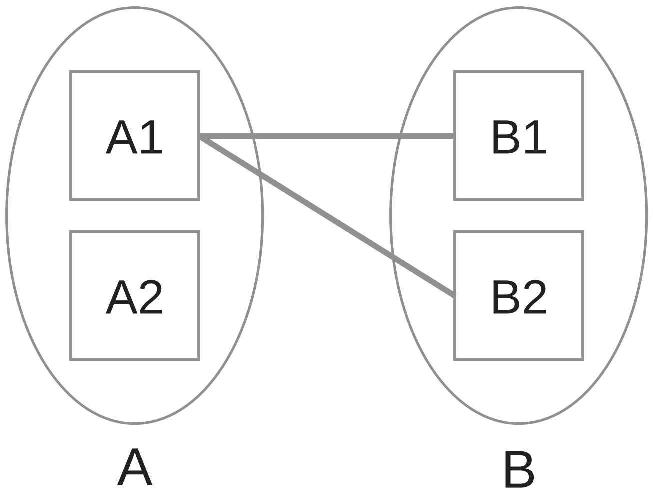


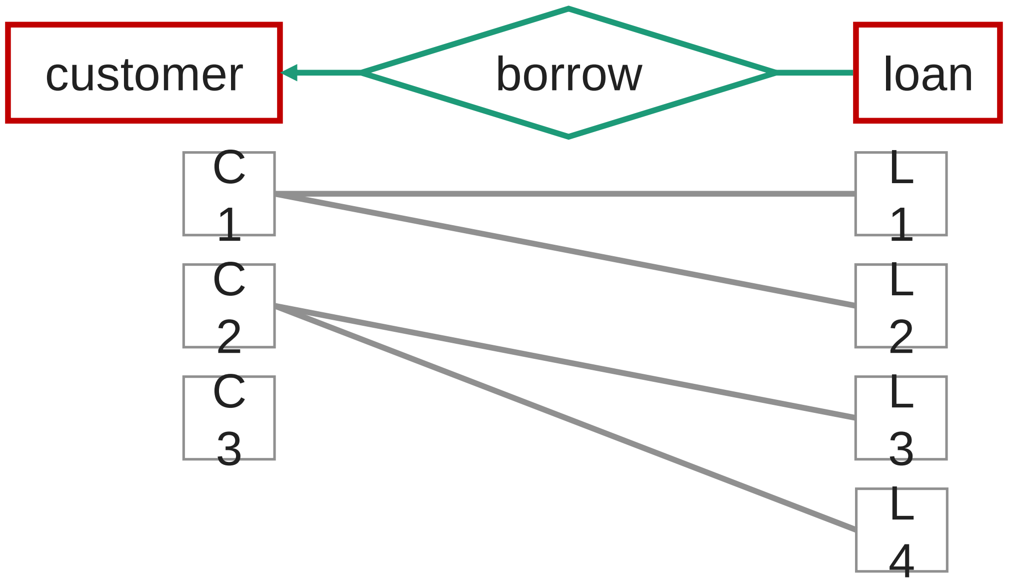


- Example: A **customer is connected with only one loan** using the relationship borrower and a **loan is connected with only one customer** using borrower.

One-to-Many relationship (1 – N)

- An entity in **A is associated with more than one entities in B** and an entity in **B is associated with only one entity in A**.

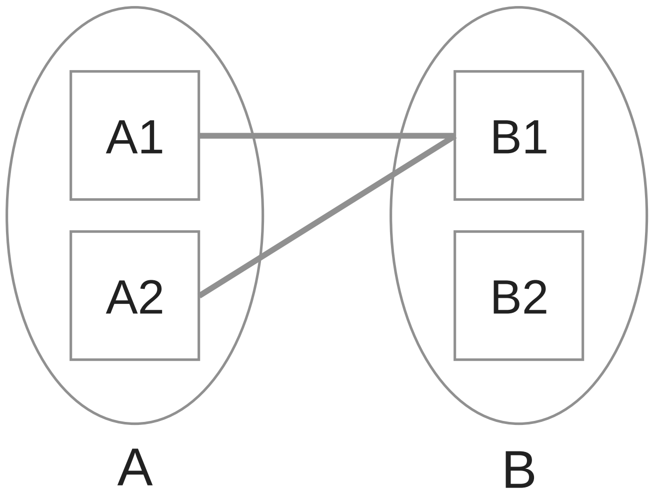


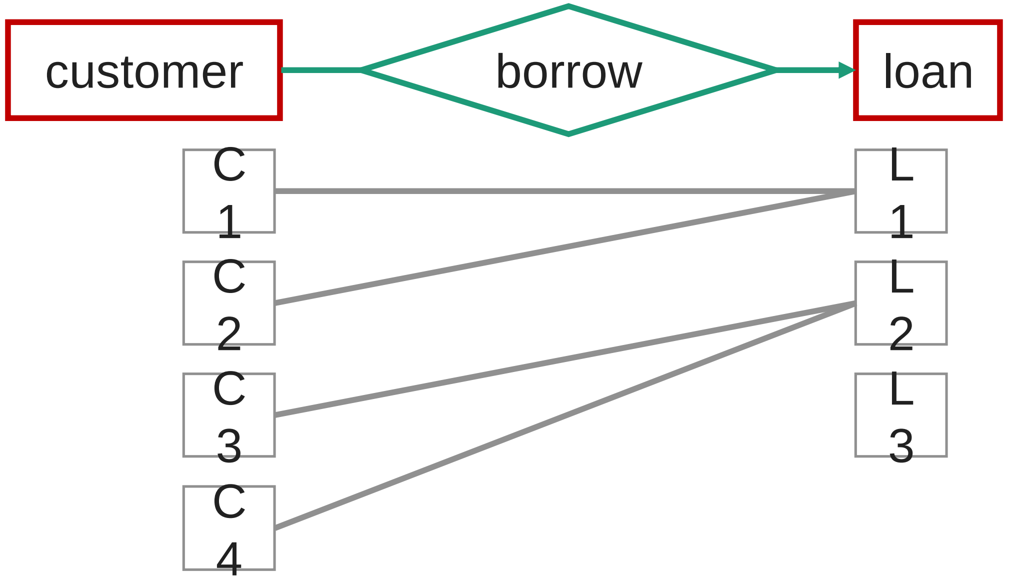


- Example: A **loan is connected with only one customer** using borrower and a **customer is connected with more than one loans using borrower**.

Many-to-One relationship (N – 1)

- An entity in **A is associated with only one entity in B** and an entity in **B is associated with more than one entities in A**.

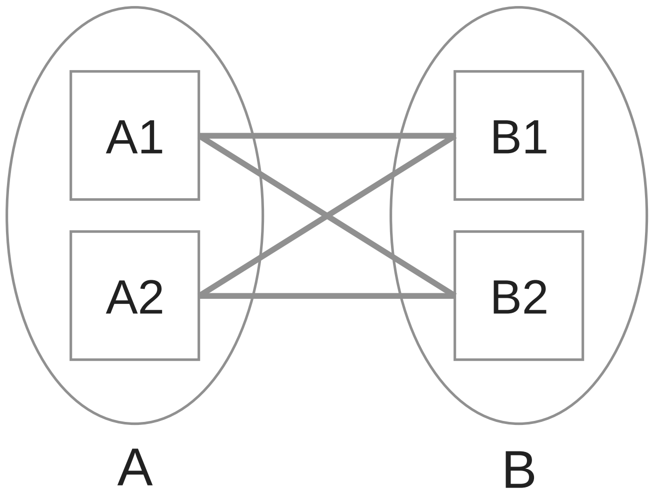


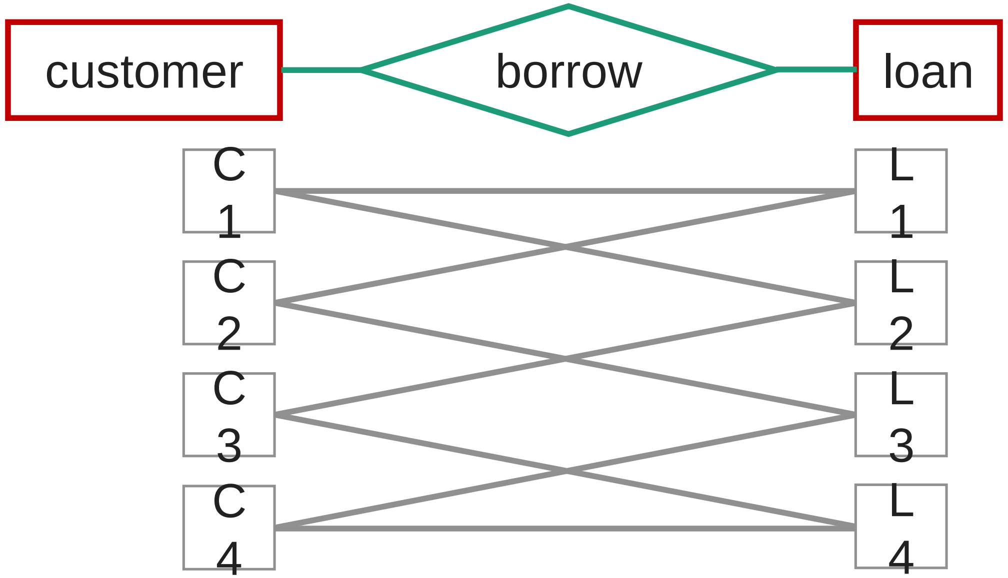


- Example: A **loan is connected with more than one customer** using borrower and a **customer is connected with only one loan** using borrower.

Many-to-Many relationship (N – N)

- An entity in **A is associated with more than one entities in B** and an entity in **B is associated with more than one entities in A**.





- Example: A **customer is connected with more than one loan** using borrower and a **loan is connected with more than one customer** using borrower.

Mapping Cardinality (Cardinality Constraints) [Exercise]

- Draw an E-R diagram and specify which type of mapping cardinality will be there in the following examples:
  - Each customer has only one account in the bank and each account is held by only one customer. [single account]
  - Each customer has only one account in the bank but an account can be held by more than one customer. [joint account]
  - A customer may have more than one account in the bank but each account is held by only one customer. [multiple accounts]
  - A customer may have more than one account in the bank and each account is held by more than one customer. [join account as well as multiple accounts]
  - A student can work in more than one project and a project can be done by more than one student.
  - A student can issue more than one book but a book is issued to only one student.
  - A subject is taught by more than one faculty and a faculty can teach more than one subject.

Participation Constraints

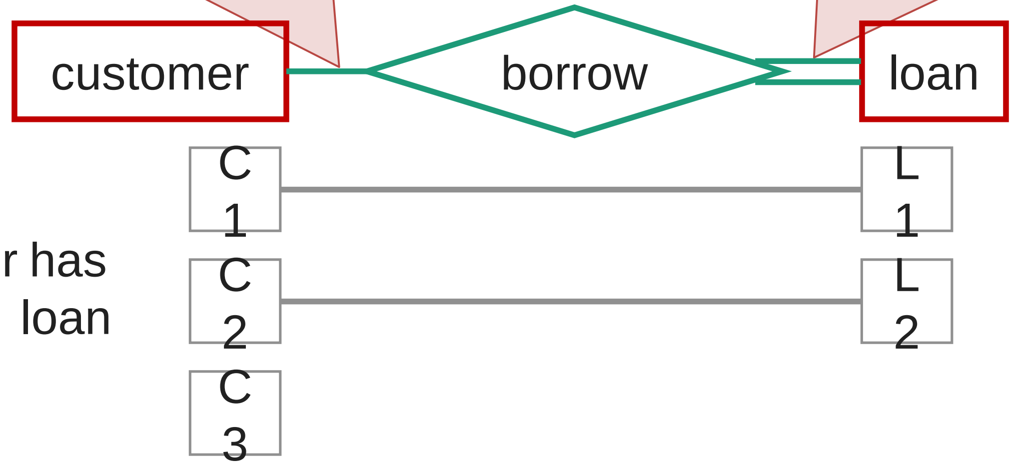
Section - 4

Participation Constraints

- It specifies the **participation of an entity set** in a relationship set.
- There are two types participation constraints
  - Total participation
  - Partial participation

Partial participation
- some entities in the entity set may not participate in any relationship in the relationship set.
- indicated by single line

Total participation
- every entity in the entity set participates in at least one relationship in the relationship set.
- indicated by double line



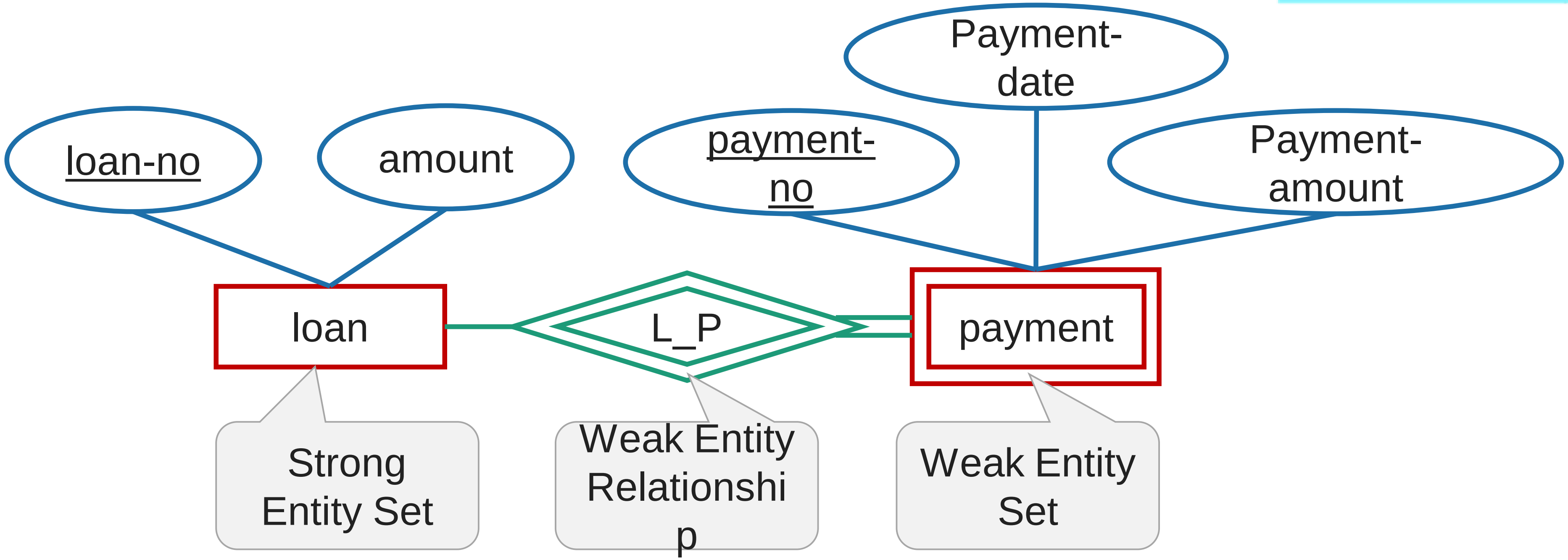
Each customer has maximum one loan

Weak Entity Set

Section - 5

Weak Entity Set

- An **entity set that does not have a primary key** is called weak ent



- Weak entity set is indicated by double rectangle.
- Weak entity relationship set is indicated by double diamond.

Weak Entity Set

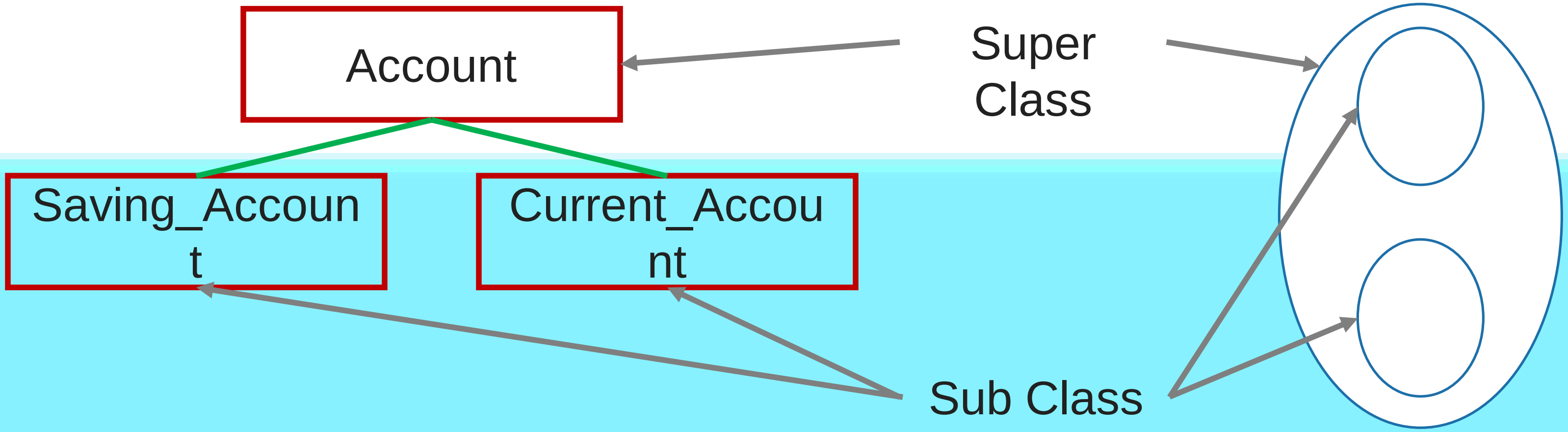
- The **existence of a weak entity set** depends on the **existence of a strong entity set**.
- The **discriminator (partial key)** of a weak entity set is the set of **attributes that distinguishes all the entities** of a weak entity set.
- The **primary key** of a weak entity set is created by **combining the primary key of the strong entity set** on which the weak entity set is existence dependent and the **weak entity set's discriminator**.
- We underline the discriminator attribute of a weak entity set with a **dashed line**.
- Payment entity has payment-no which is discriminator.
- Loan entity has loan-no as primary key.
- So primary key for payment is **(loan-no, payment-no).**

Superclass v/s Subclass

Section - 6

Superclass v/s Subclass

Super Class	Sub Class
A superclass is an entity from which **another entities can be derived**.	A subclass is an entity that is **derived from another entity**.
E.g, an entity account has two subsets saving_account and current_account So an **account is superclass**.	E.g, saving_account and current_account entities are derived from entity account. So **saving_account and current_account are subclass**.

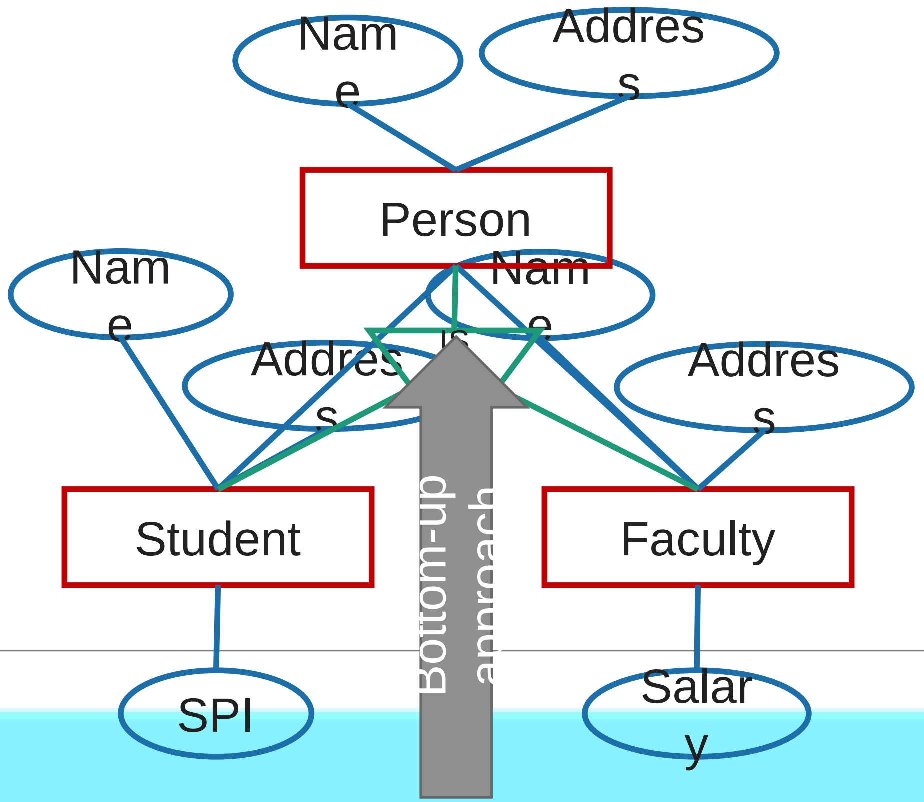


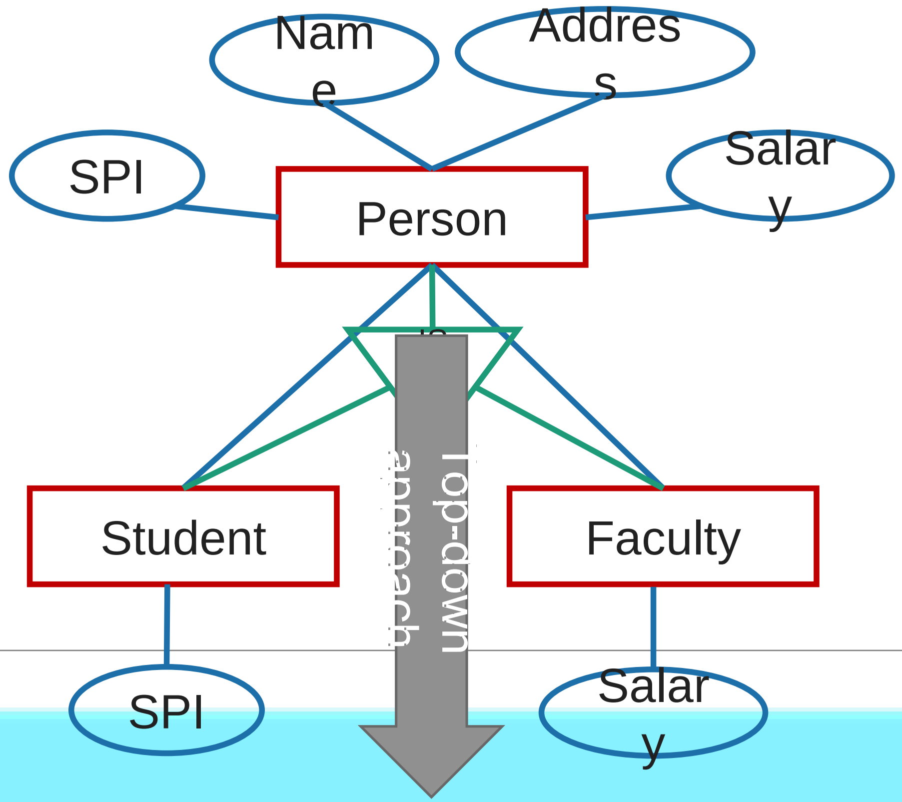
Generalization v/s Specialization

Section - 7

Generalization v/s Specialization

Generalization	Specialization
It **extracts the common features** of **multiple entities** to **form a new entity**	It **splits an entity to form multiple new entities** that **inherit some feature of the**

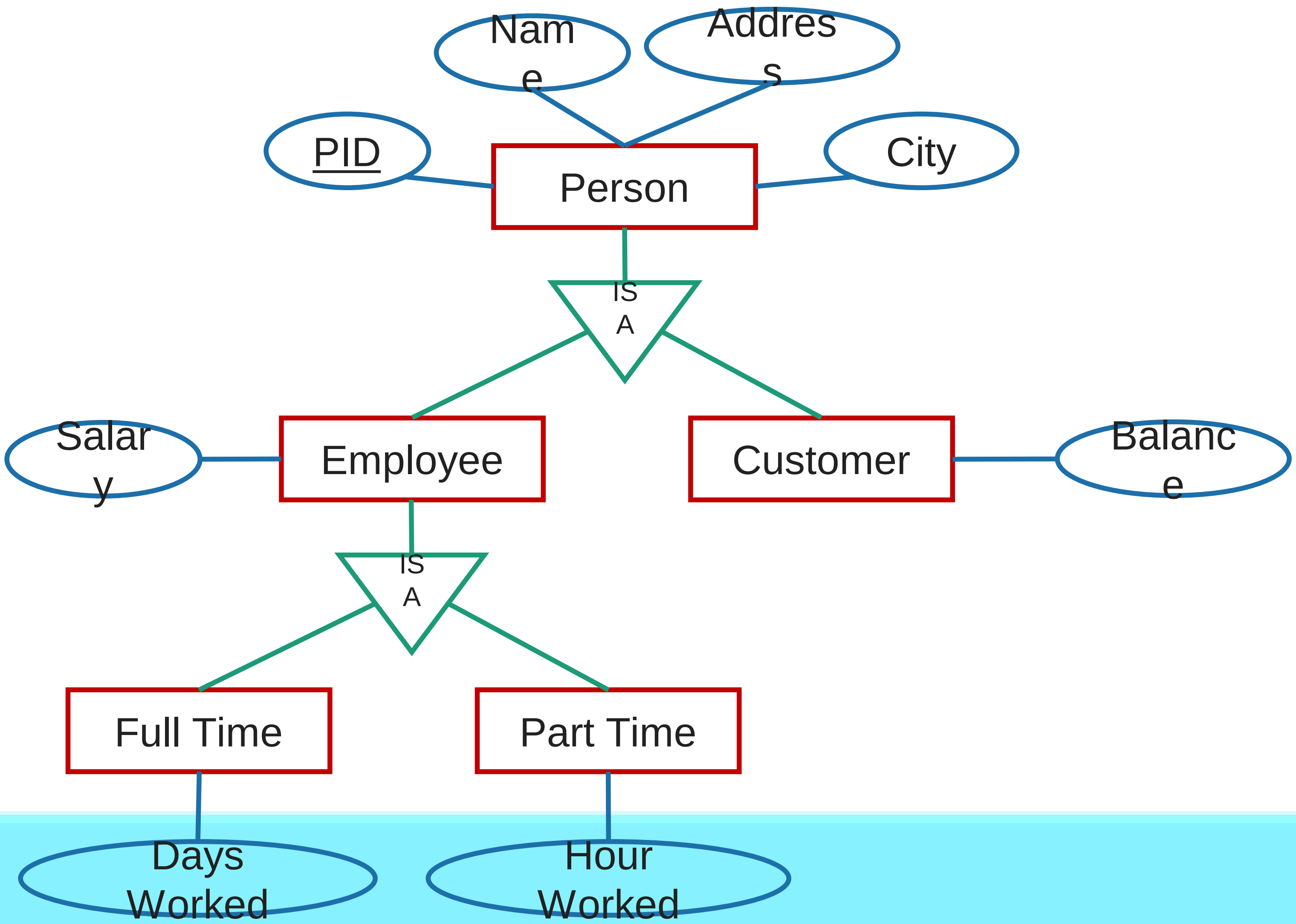




Generalization v/s Specialization

Generalization	Specialization
The process of **creation of group from various entities** is called generalization.	The process of **creation of sub-groups within an entity** is called specialization.
It is **Bottom-up** approach.	It is **Top-down** approach.
The process of taking the **union of two or more lower level entity** sets to produce a higher level entity set.	The process of taking a **sub set of higher level entity set** to form a lower level entity set.
It starts from the number of entity sets and creates high level entity set using some common features.	It starts from a single entity set and creates different low level entity sets using some different features.

Generalization & Specialization example



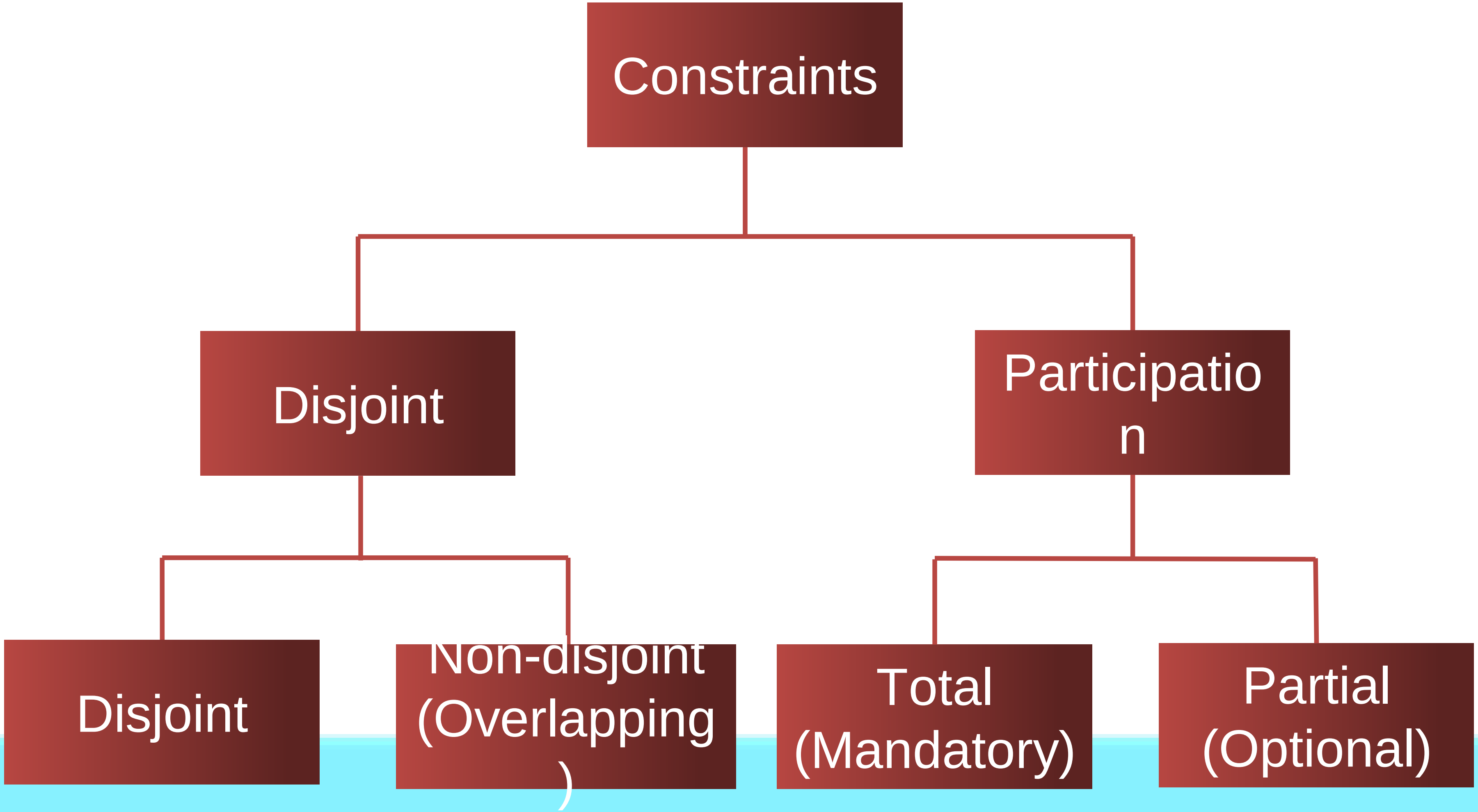
Exercise

- Give the examples of Generalization/Specialization in the following E-R diagram:
  - Hospital Management System.
  - College Management System.
  - Bank Management System.
  - Insurance Company.

Constraints on Specialization and Generalization

Section - 8

Constraints on Specialization and Generalization



Disjoint Constraint

- It describes **relationship between members of the superclass and subclass** and indicates whether member of a superclass can be a member of one, or more than one subclass.
- Types of disjoint constraints
  - Disjoint Constraint
  - Non-disjoint (Overlapping) Constraint

Disjoint Constraint

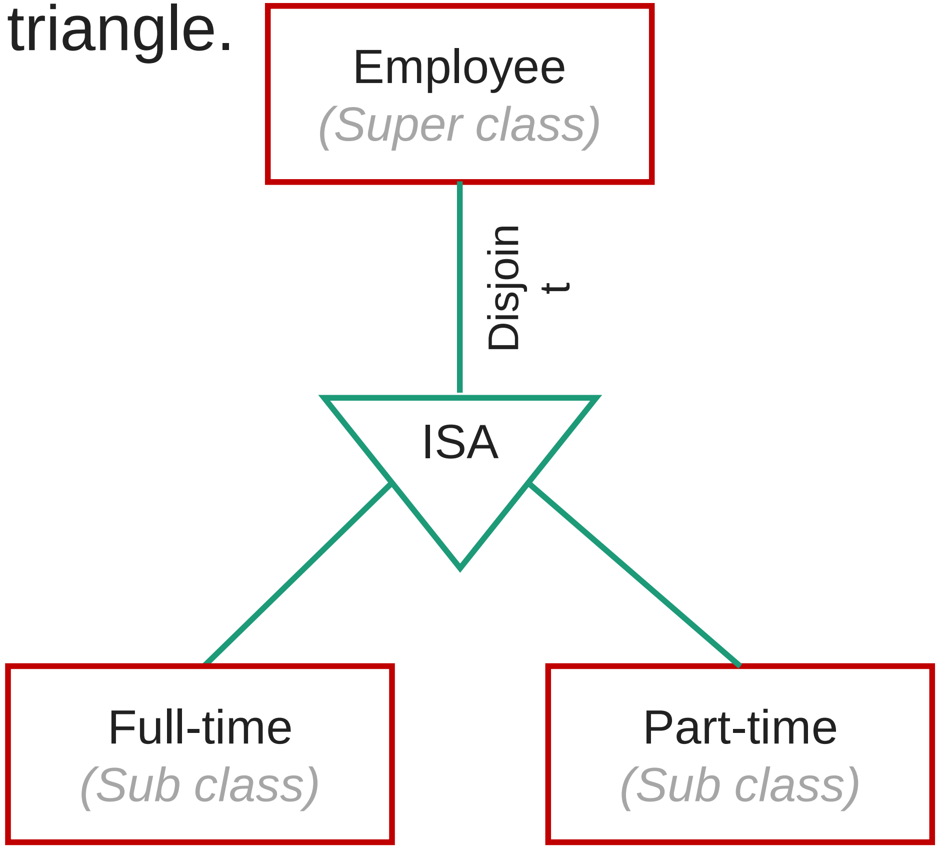
- It specifies that the **entity of a super class can belong to only one lower-level entity set** (sub class).
- Specified by ‘**d**’ or by writing **disjoint** near to the ISA triangle.


Cricketer
(Super class)

Batsman
(Sub class)

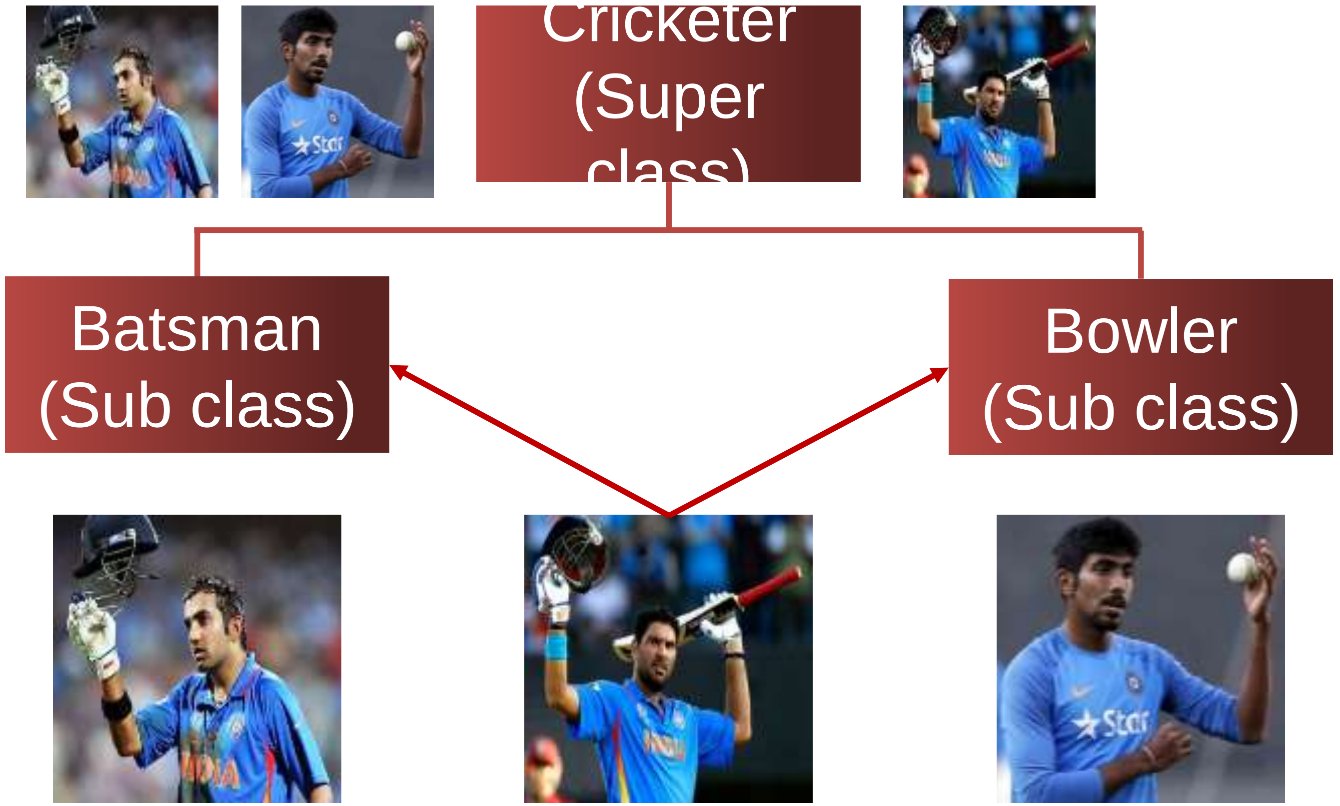
Bowler
(Sub class)


All the players are associated with only one sub class either (Batsman or Bowler).

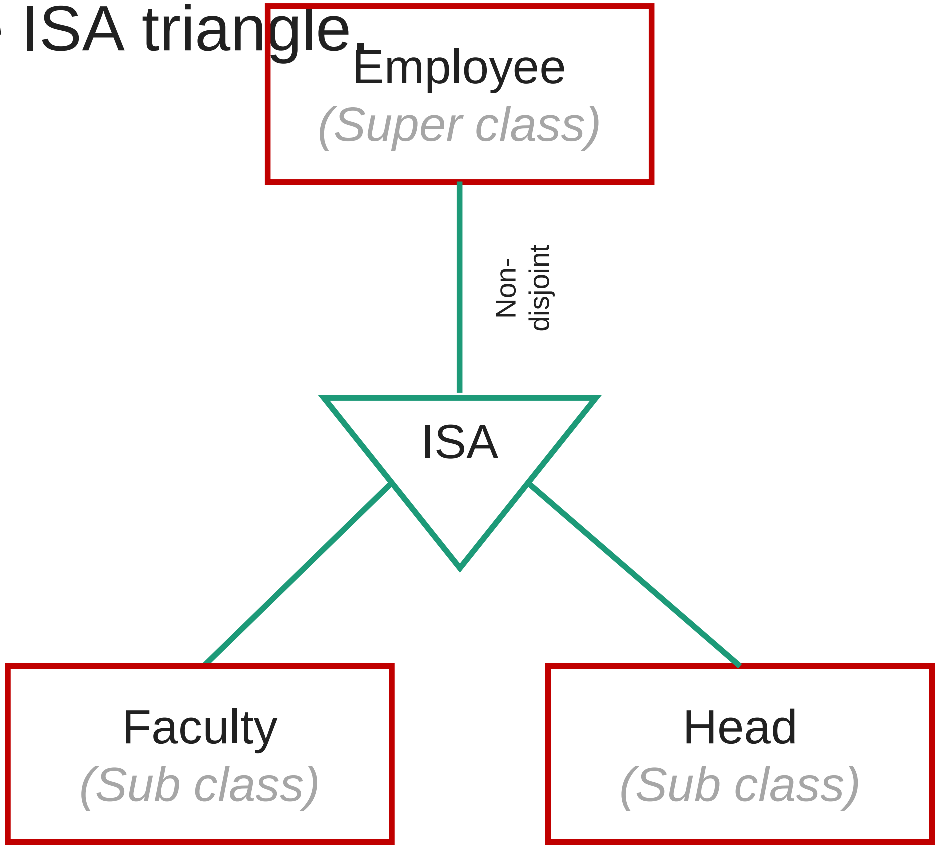


Non-disjoint (Overlapping) Constraint

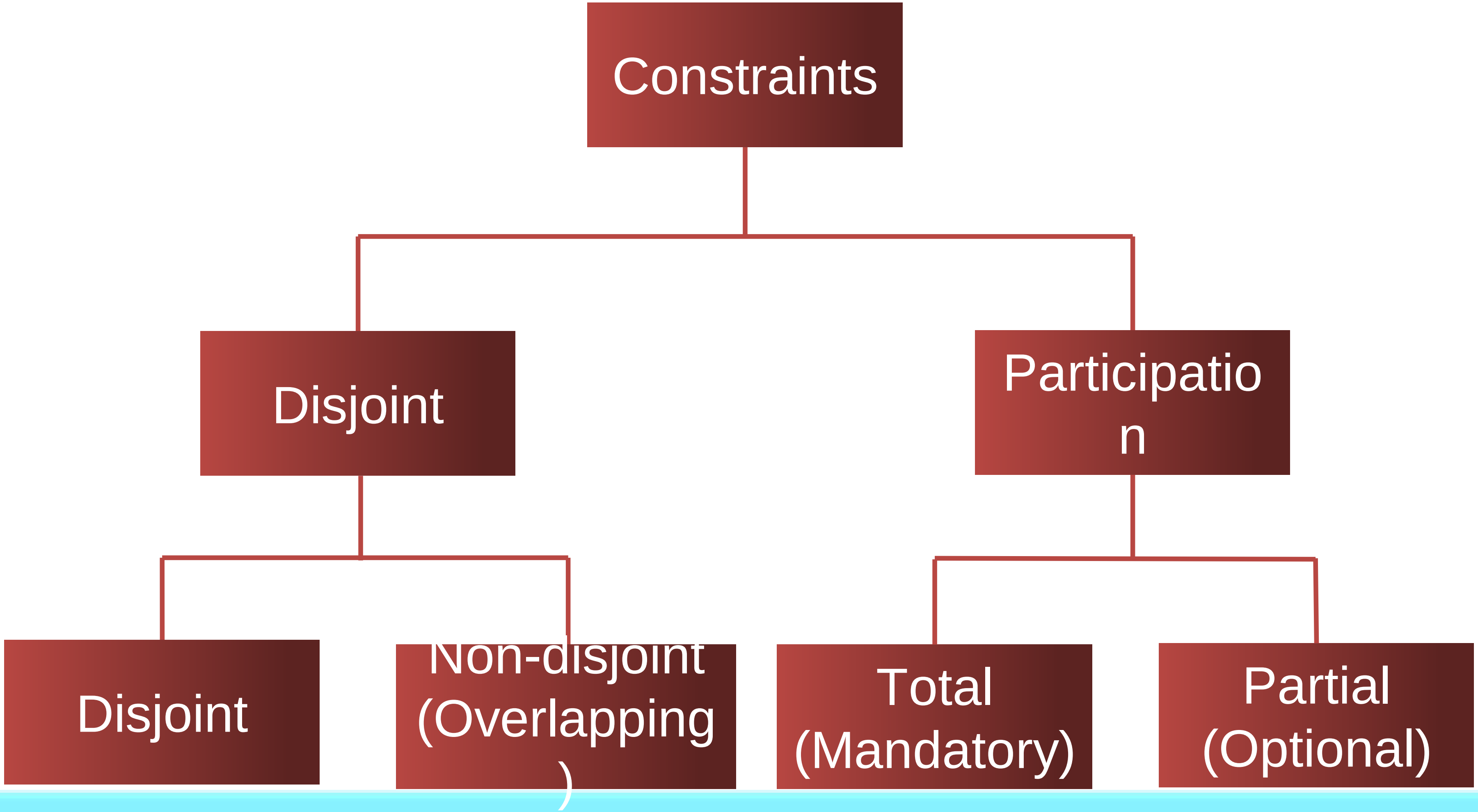
- It specifies that an **entity of a super class can belong to more than one lower-level entity** set (sub class).
- Specified by '**o**' or by writing **overlapping** near to the ISA triangle.



One player (Yuvraj singh) is associated with more than one sub class.



Constraints on Specialization and Generalization



Participation (Completeness) Constraint

- It determines **whether every member of super class must participate as a member of subclass or not**.
- Types of participation (Completeness) Constraint
  - Total (Mandatory) participation
  - Partial (Optional) participation

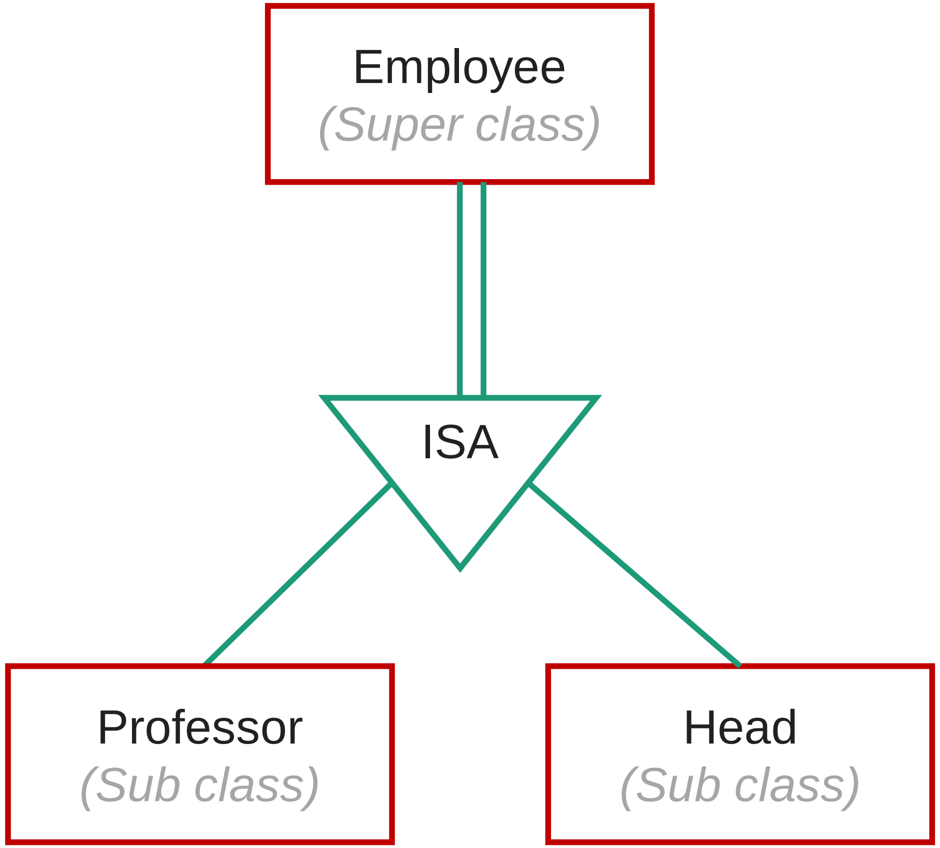
Total (Mandatory) Participation

- Total participation specifies that **every entity in the superclass must be a member of some subclass** in the specialization.
- Specified by a **double line** in E-R diagram.


Cricketer (Super class)

Batsman (Sub class)

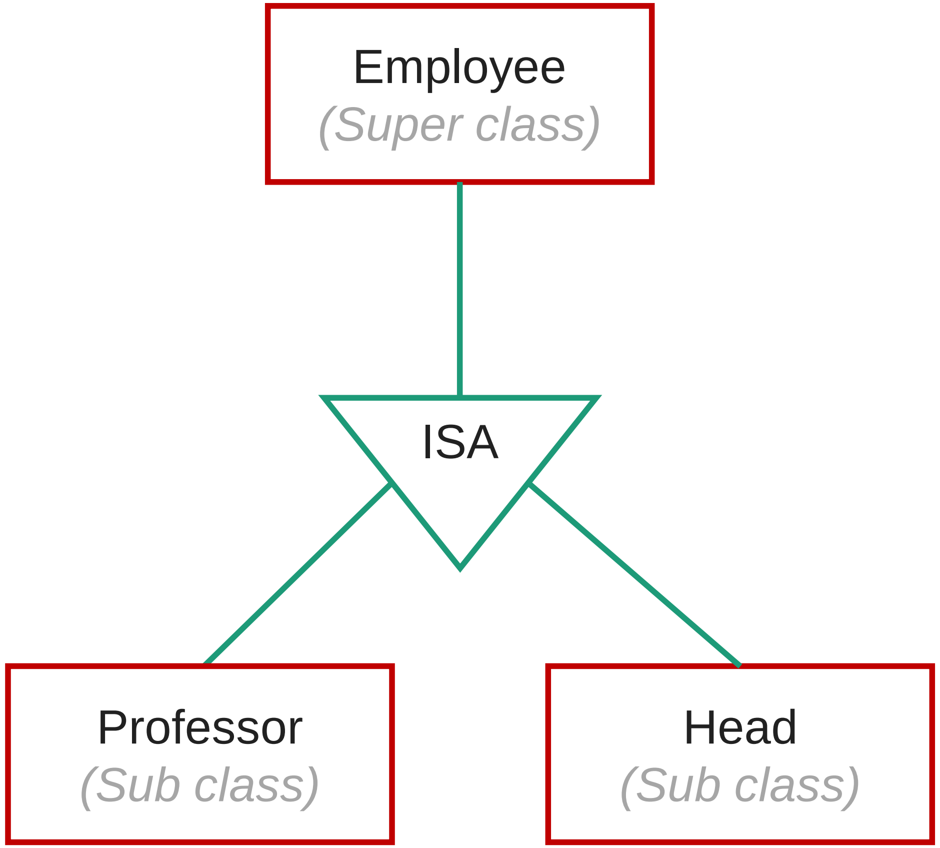
Bowler (Sub class)

All the players are associated with minimum one sub class either (Batsman or Bowler).

Partial (Optional) Participation

- Partial participation specifies that **every entity in the super class does not belong to any of the subclass** of specialization.
- Specified by a **single line** in E-R diagram.

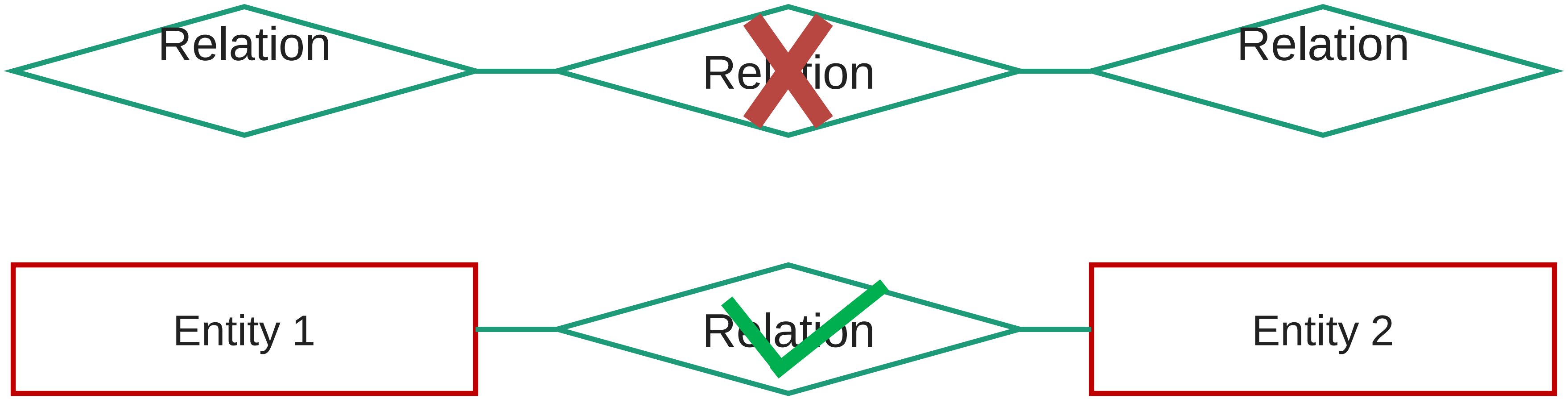


Aggregation in E-R diagram

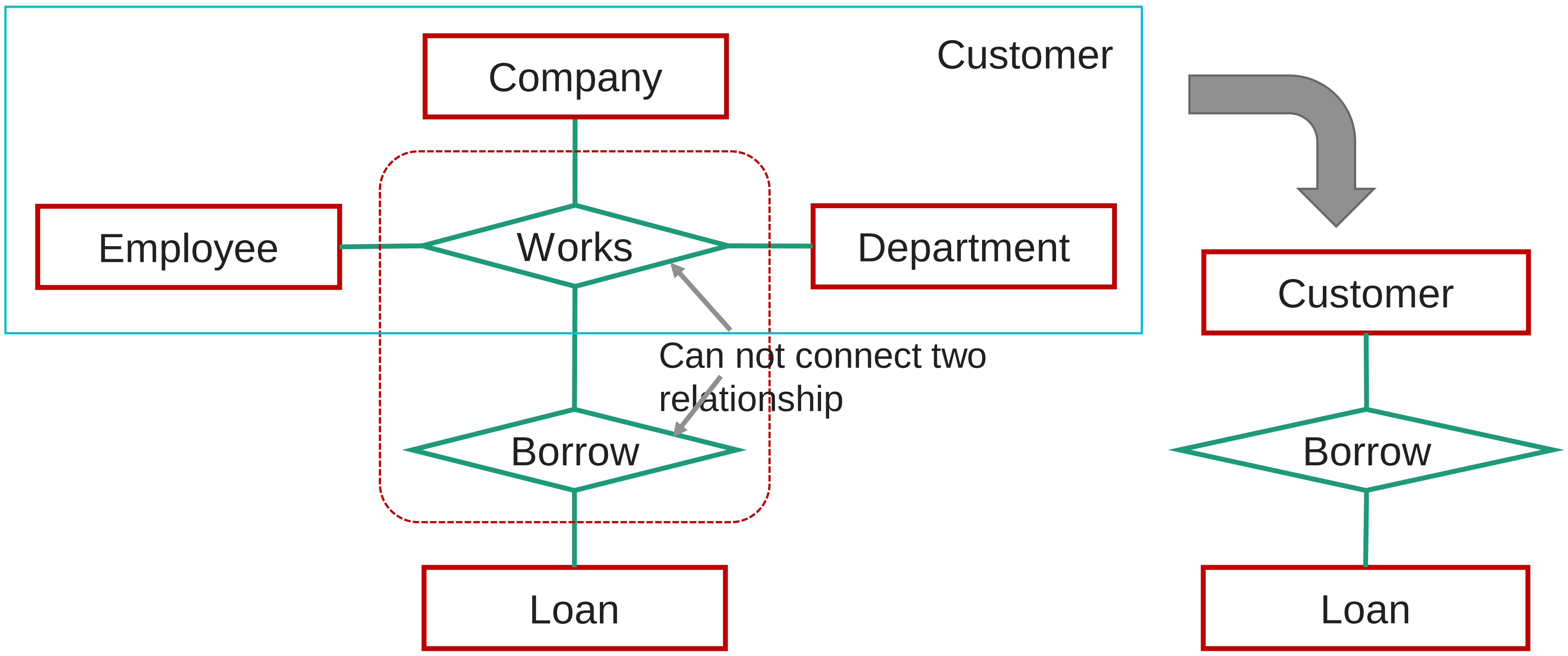
Section - 9

Limitation of E-R diagram

- In E-R model we **cannot express relationships between two relationships**.



Limitation of E-R diagram

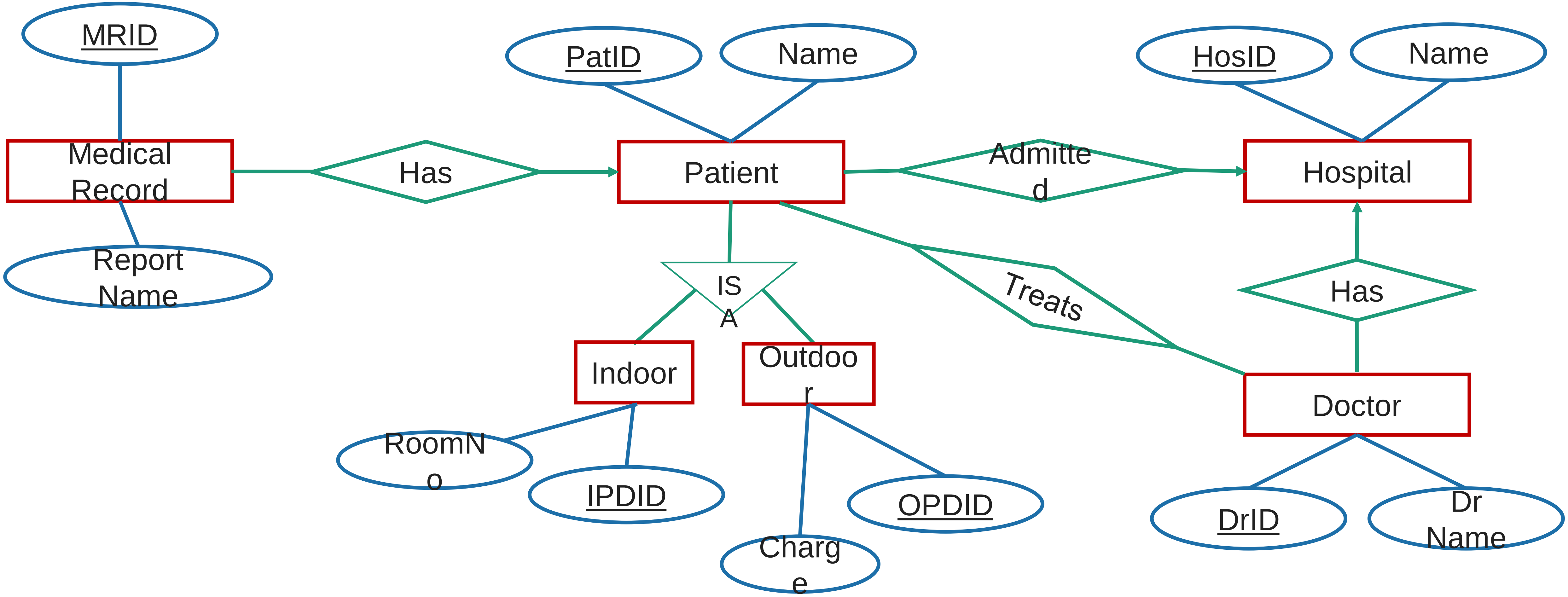


Process of creating an entity by combining various components of E-R diagram is called aggregation.

E-R diagram of Hospital Management System

Section - 10

E-R diagram of Hospital Management System



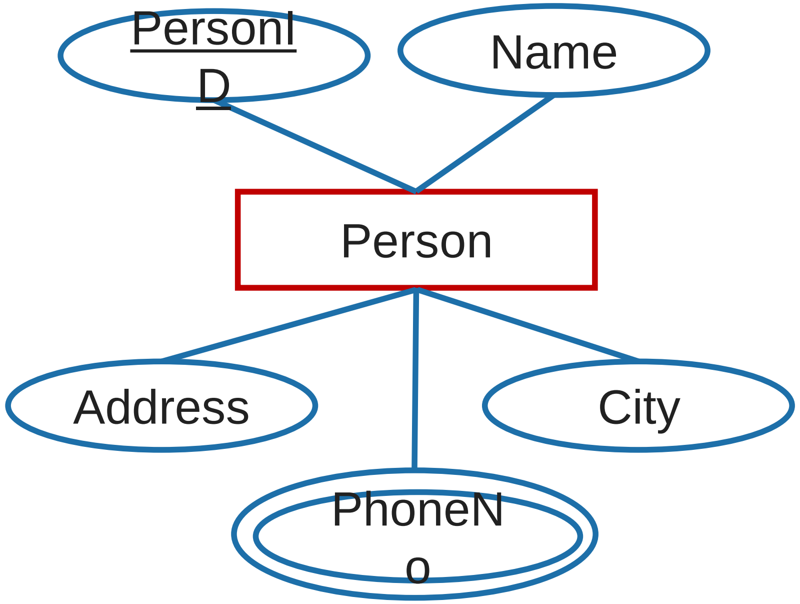
Reduce the E-R diagram to Database Schema

Section - 11

Reduce the E-R diagram to database schema

Step 1: Reduce **Entities** and **Simple Attributes**:

- An **entity** of an ER diagram is **turned into** a **table**.
- Each **attribute** (except multi-valued attribute) **turns into** a **column** (attribute) in the table.
- **Table name** can be same as **entity name**.
- **Key attribute** of the entity is the **primary key** of the table which is usually underlined.
- It is highly recommended that every table should start with its primary key attribute conventionally named as TablenameID.

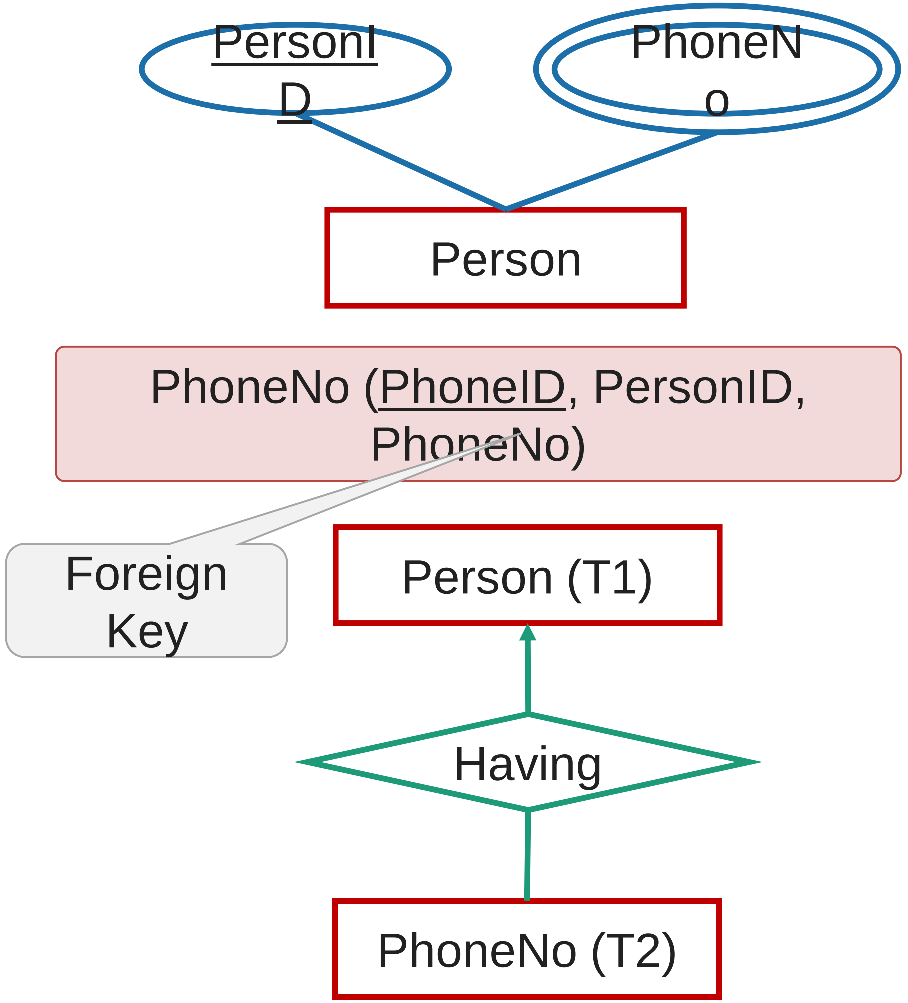


Person (PersonID, Name, Address, City)

Reduce the E-R diagram to database schema

Step 2: Reduce **Multi-valued Attributes**:

- **Multi-value attribute** is turned into a **new table**.
- **Add** the **primary key** column into **multi-value attribute's table**.
- Add the **primary key column** of the **parent entity's table** as a **foreign key** within the **new (multi-value attribute's) table**.
- Then make a **1:N relationship** between the Person table and PhoneNo table.



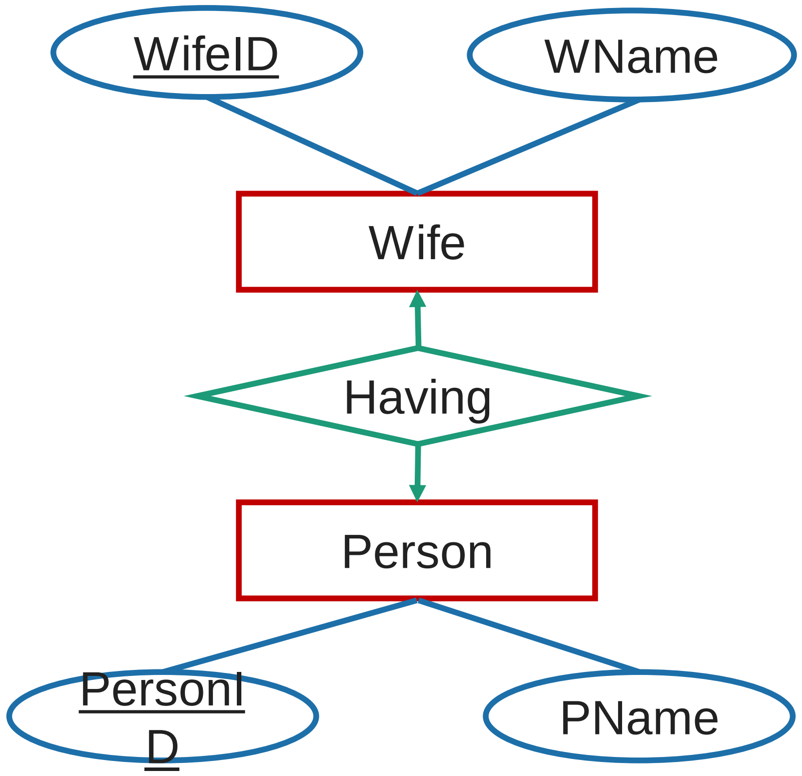
Reduce the E-R diagram to database schema

Step 3: Reduce **1:1 Mapping Cardinality**:

- Convert **both entities** in to **table** with proper attribute.
- Place the **primary key** of any **one table** in to the **another table** as a **foreign key**.
- Place the primary key of the Wife table WifeID in the table Persons as Foreign key.

OR

- Place the primary key of the Person table PersonID in the table Wife as Foreign key.



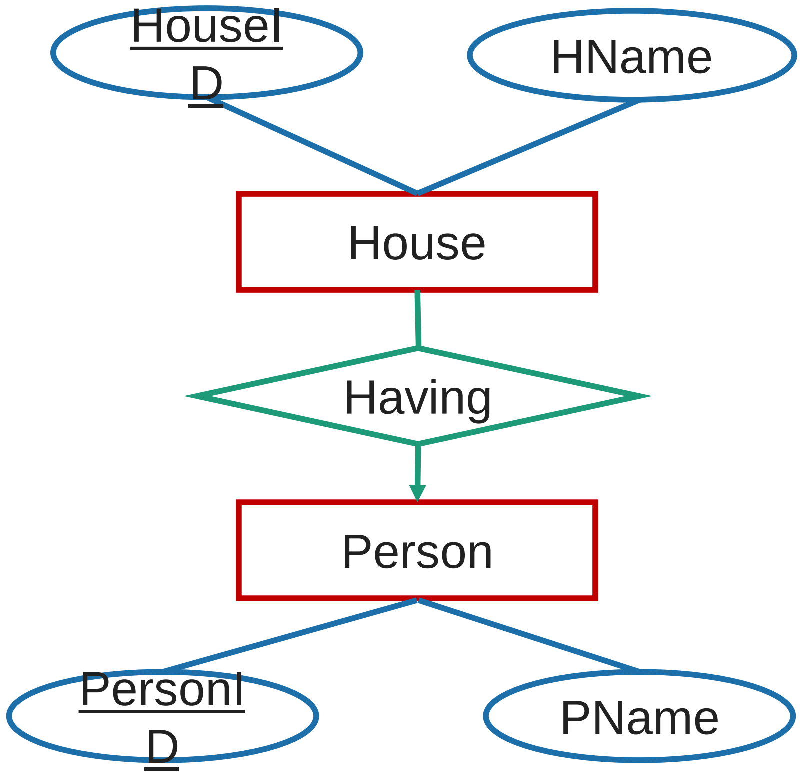
Person (<u>PersonID</u>, PName)
Wife (<u>WifeID</u>, Wname, PersonID)

Wife (<u>WifeID</u>, Wname)
Person (<u>PersonID</u>, Pname, WifeID)

Reduce the E-R diagram to database schema

Step 4: Reduce **1:N Mapping Cardinality**:

- Convert **both entities** in to **table** with proper attribute.
- Place the **primary key** of **table having 1 mapping** in to the another **table having many cardinality as a Foreign key**.
- Place the primary key of the Person table PersonID in the table House as Foreign key.

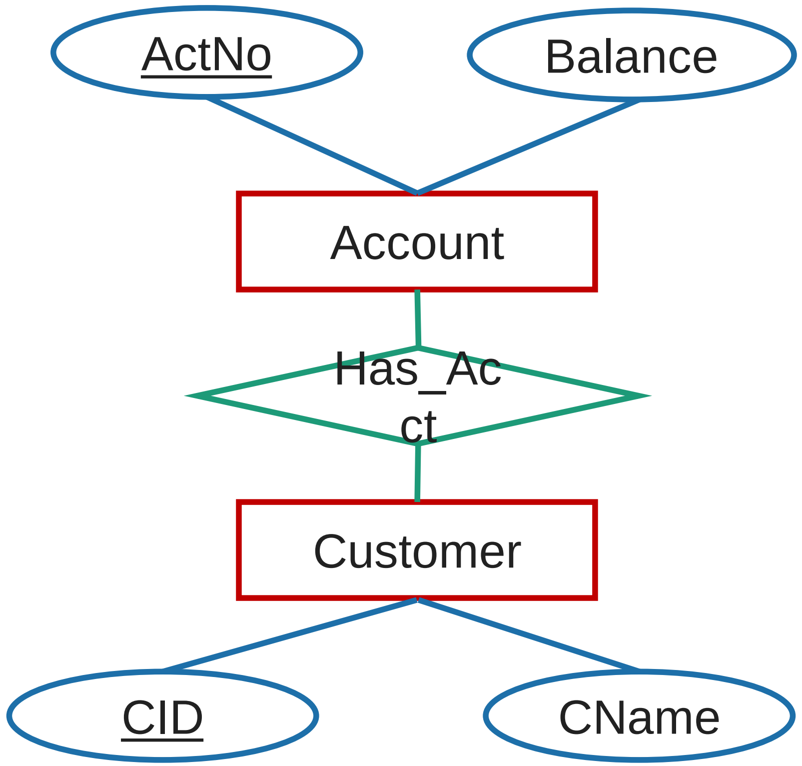


Person (<u>PersonID</u>, PName)
House (<u>HouseID</u>, Hname, PersonID)

Reduce the E-R diagram to database schema

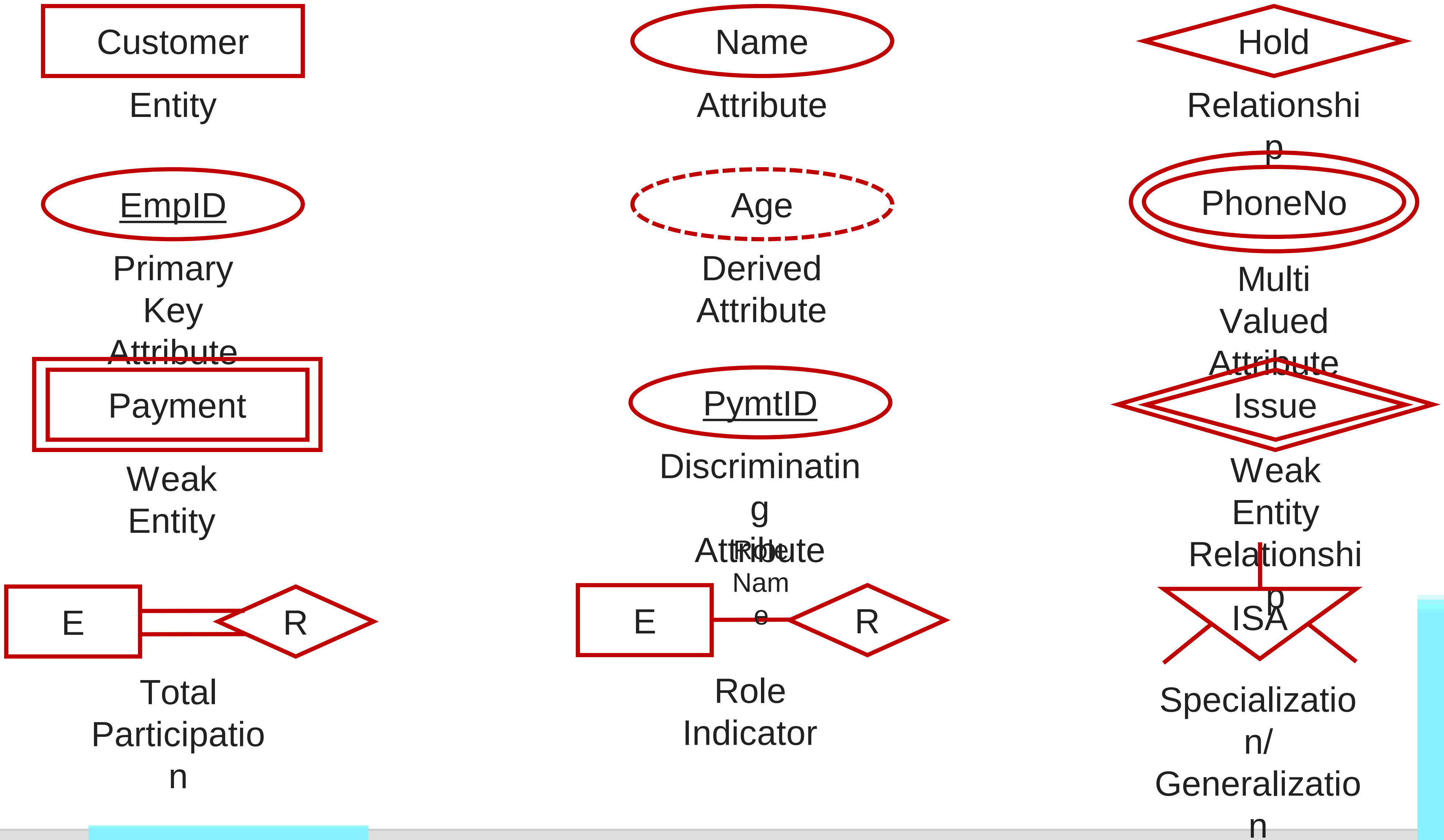
Step 5: Reduce **N:N Mapping Cardinality**:

- Convert both **entities** in to **table** with proper attribute.
- Create a **separate table for relationship**.
- Place the **primary key of both entities table** into the **relationship's table as foreign key**.
- Place the primary key of the Customer table CID and Account table Ano in the table Has_Acct as Foreign key.

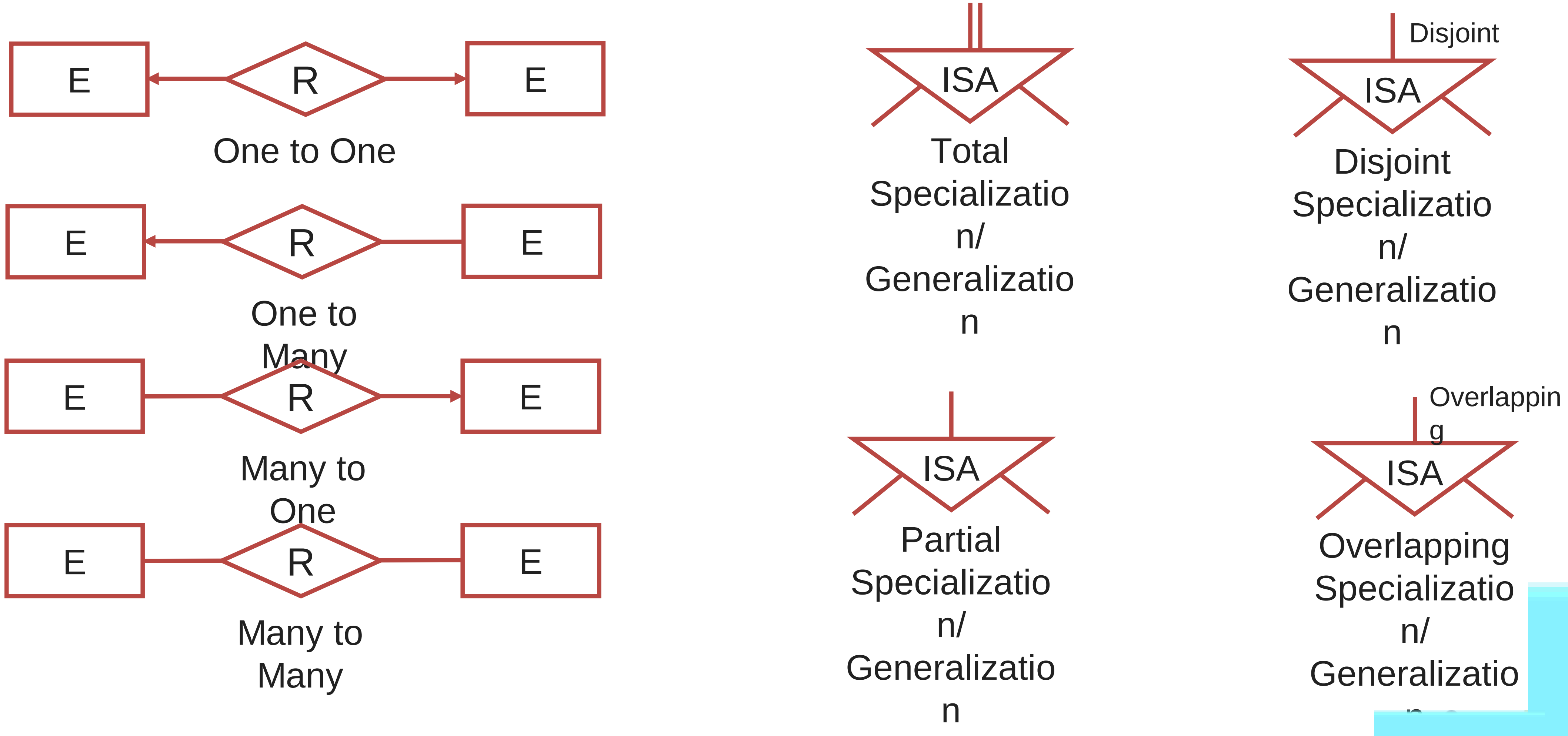


Customer (CID, CName)
Account (ActNo, Balance)
Has_Acct (HasAcctID, CID, ActNo)

Summery of Symbols used in E-R diagram



Summery of Symbols used in E-R diagram



Data Models

Section - 12

What is a Database Models?

- A database model is a type of data model that **defines the logical structure of a database**.
- It determine **how data can be stored, accessed and updated** in a database management system.
- The most popular example of a database model is the relational model, which uses a table-based format.

Type of Database Models

- Hierarchical Model
- Network Model
- Entity-relationship Model
- Relational Model
- Object-oriented database Model

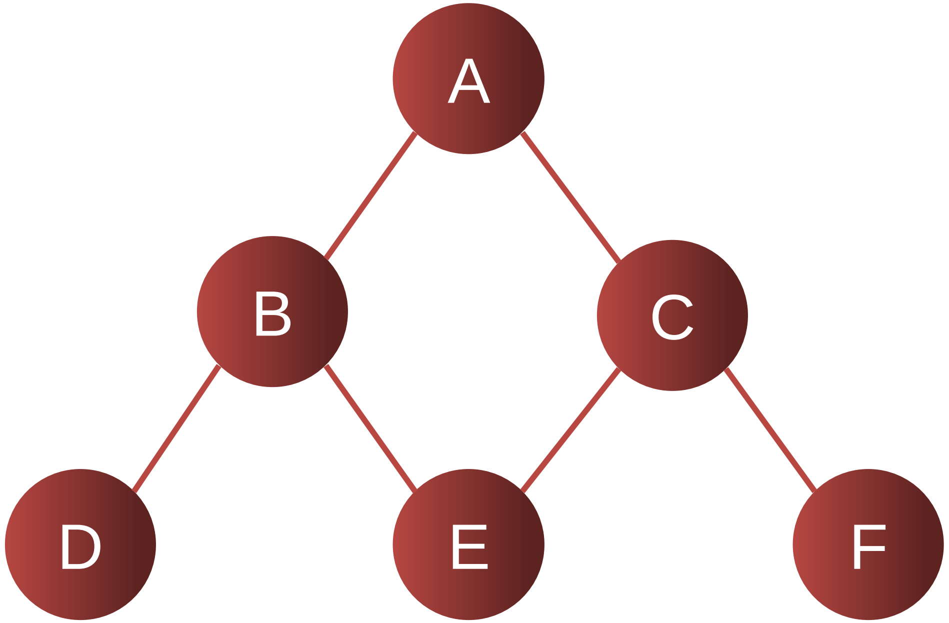
Hierarchical Model

- The hierarchical model organizes data into a **tree-like structure**, where **each record has a single parent or root**.

- The hierarchy **starts from the Root data**, and **expands like a tree**, **adding child nodes to the parent nodes**.
- In hierarchical model, data is organized into **tree-like structure** with **one-to-many relationship** between two different types of data, for example, **one department can have many professors and many students**.

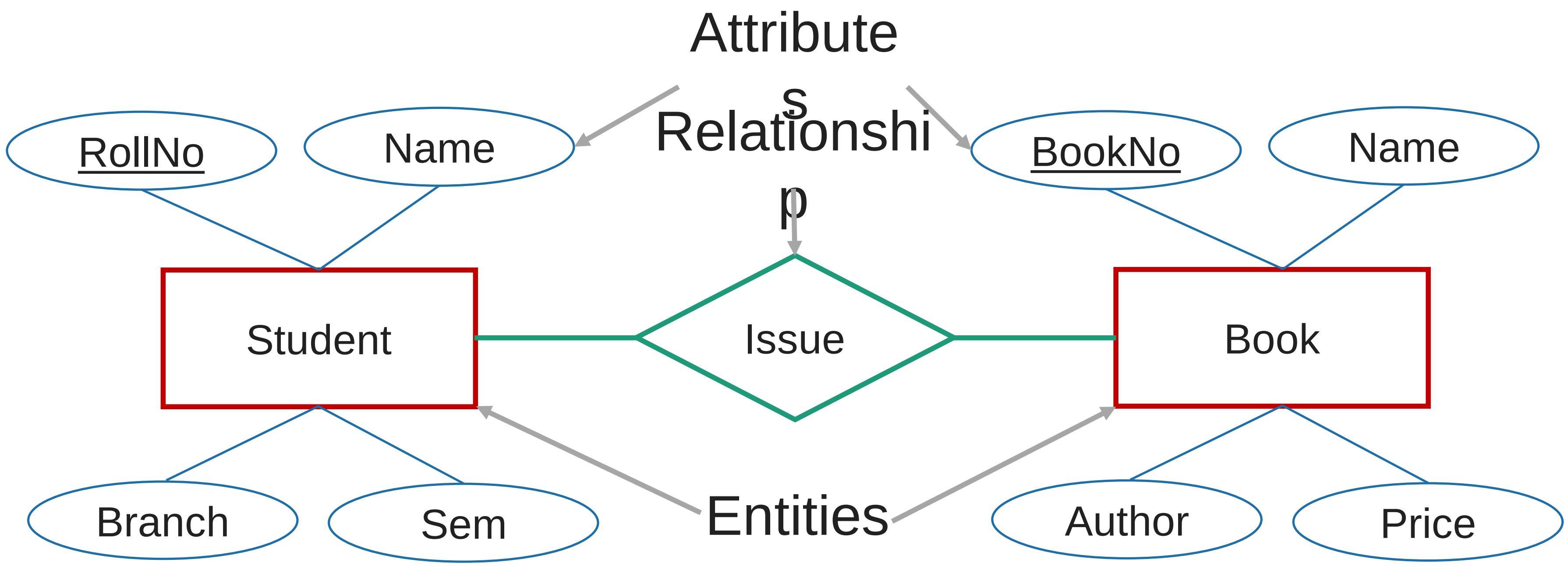
Network Model

- This is an **extension of the hierarchical model**, allowing **many-to-many relationships** in a tree-like structure that **allows multiple parents**.



Entity-relationship Model

- In this database model, **relationships are created by dividing object of interest into entity** and **its characteristics into attributes**.



Relational Model

- In this model, **data is organized in two-dimensional tables** and the **relationship is maintained by storing a common attribute**.

Rno	Student_Name	Age
101	Raj Patel	20
102	Meet Shah	21

SubID	Subject_Name	Teacher
1	DBMS	Doshi
2	DS	Vyash

Foreign Key

Foreign Key

ResID	**Rno**	**SubID**	Marks
1	101	1	80
2	101	2	85
3	102	1	75
4	102	2	80

Object-oriented database Model

- This data model is another method of representing real world objects.
- It considers **each object in the world as objects** and isolates it from each other.
- It **groups its related functionalities together** and **allows inheriting its functionality** to other related sub-groups.

Integrity Constraints

Section - 13

Integrity Constraints

- Integrity constraints are a **set of rules**. It is used to **maintain the quality** of information.
- Integrity constraints ensure that the data insertion, updating, and other processes have to be performed in such a way that data integrity is not affected.
- Thus, integrity constraint is used to **guard against accidental damage** to the database.
- Various Integrity Constraints are:
  - Check
  - Not null
  - Unique
  - Primary key
  - Foreign key

Integrity Constraints

- Check
  - This constraint defines a business rule on a column. All the rows in that column must satisfy this rule.
  - Limits the data values of variables to a **specific set, range, or list of values**.
  - The constraint can be applied for a single column or a group of columns.
  - E.g. value of SPI should be between 0 to 10.
- Not null
  - This constraint ensures all rows in the table contain a definite value for the column which is specified as not null. Which means a **null value** is not allowed.
  - E.g. name column should have some value.
- Unique
  - This constraint ensures that a column or a group of columns in each row have a **distinct (unique)** value.
  - A column(s) can have a null value but the values cannot be duplicated.
  - E.g. enrollmentno column should have unique value.

Integrity Constraints

- Primary key
  - This constraint defines a column or combination of columns which uniquely identifies each row in the table.
  - Primary key = **Unique key + Not null**
  - E.g. enrollmentno column should have unique value as well as can’t be null.
- Foreign key (referential integrity constraint)
  - A referential integrity constraint (foreign key) is specified between two tables.
  - In the referential integrity constraints, if a foreign key column in table 1 refers to the primary key column of table 2, then every value of the foreign key column in table 1 must be null or be available in primary key column of table 2.

Foreign Key

DeptID	Dept_Name	HOD
1	Computer	Doshi
2	IT	Vyash

RollNo	**Student_Name**	**DeptID**
101	Raj Patel	1
102	Meet Shah	2

Questions asked in GTU

1. Write a note on mapping cardinality in E-R diagram.
2. Explain the difference between a weak and a strong entity set.
3. Explain the difference between generalization and specialization. **OR** Explain specialization and generalization concept in E-R diagram with suitable example.
4. Write a note on constraints on specialization and generalization.
5. Explain aggregation in E-R diagram with example.
6. What do you mean by integrity constraints? Discuss various integrity constraints.

Questions asked in GTU [E-R diagrams]

7. Draw E-R diagram for Bank Management System.
8. Define E-R diagram. Draw an E-R diagram for Library Management System. Assume relevant entities and attributes for the given system.
9. Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.
10. Design a generalization–specialization hierarchy for a motor-vehicle sales company. The company sells motorcycles, passenger cars, vans, and buses. Justify your placement of attributes at each level of the hierarchy. Explain why they should not be placed at a higher or lower level.

Questions asked in GTU [E-R diagrams and Database]

11. Design a database for an airline. The database must keep track of customers and their reservations, flights and their status, seat assignments on individual flights, and the schedule and routing of future flights. Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key constraints.

12. Design a database for a hospital with a set of patients and a set of medical doctors. Associate with each patient a log of the various tests and examinations conducted. Your design should include an E-R diagram, a set of relational schemas, and a list of constraints, including primary-key and foreign-key constraints.

Thank You